# THE AREA FUNCTION OF THE MERTENS EXCURSION SKELETON AND HORIZON-LOCALIZED CRITERIA FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. We decompose the Mertens function $M(x) = \sum_{n \le x} \mu(n)$ into its excursions between zero runs and study a regularized triangular skeleton of these excursions, its area curves, and exact quadratic and correlation identities inside each excursion. Two elementary counting lemmas show that, among excursions starting at most $X$, those with $x_1 > x_0^2$ number $O(\log\log X)$, and those with $x_1 > \lambda x_0$ number $O(\log X/\log\lambda)$. Combined with the Kaczorowski–Pintz lower bound for large alternating oscillations, this yields a general localization theorem: the Riemann hypothesis is equivalent to Littlewood's bound $M(t) \ll_\varepsilon t^{1/2+\varepsilon}$ restricted to the points $t$ of each excursion lying below a horizon $h(x_0)$, whenever the excursions crossing the horizon are fewer than $(\gamma_1/\pi)\log X$. The horizon can be lowered from $x_0^2$ through $cx_0^2$ and $x_0\psi(x_0)$ with $\psi \to \infty$ down to $\lambda x_0$ for every $\lambda > e^{\pi/\gamma_1}$. On the resulting window $(x_0, \frac{3}{2}x_0]$ the Möbius function is linear in the prefix $\mu\mathbf{1}_{[1,x_0]}$, and RH becomes a statement about an explicit linear sum at the start of each excursion. Independently, the same counting bounds convert any uniform sub-Gaussian tail for a normalized excursion statistic into extreme scales $\sqrt{\log\log X}$ and $\sqrt{\log\log\log X}$ for the linear and square families respectively. We also give an exact bilinear representation of $M$ on the square range of an excursion, an exact Mellin decomposition of $1/\zeta$ into excursion blocks, and an asymptotic analysis of the split of $M$ into smooth and large-prime parts.

## CONTENTS

1. Introduction 2
Main results 2
Conventions 3
2. Excursions and the regularized skeleton 3
3. Exact formulas 4
4. Internal correlations of an excursion 5
5. Where $\zeta$ enters 7
6. Bounds for the area curves 8
7. Global criteria equivalent to RH 9
8. Lacunary excursion families and horizon-localized criteria 11
8.1. Geometric counts 11
8.2. Arithmetic input: Kaczorowski–Pintz 12
8.3. Horizon localization 13
8.4. The linear window and the one-step recursion 14
8.5. Conditional extreme-value consequence of excursion sparsity 17
8.6. The square range as an exact bilinear form of the prefix 18
8.7. Mellin mass of a complete excursion 19
9. Trial-division horizon of an excursion 19
10. Large-prime decomposition at level $u^2$ 20
11. Local large-prime decomposition of an excursion 22

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12. A one-frequency model for the black curve 23
13. Numerical data 24
14. Open problems 24
References 25

## 1. Introduction

The object of this paper is the Mertens function $M$, decomposed into excursions between zero runs (Section 2). Two kinds of information about $M$ are separated throughout: what is forced by the ordering and local geometry of the excursions alone, and what requires arithmetic input.

**Main results.** The central results are in Section 8.

*Geometric counts.* For any ordered family of disjoint integer intervals $(x_0, x_1]$, the members with $x_1 > x_0^A$ $(A > 1)$ starting below $X$ are $O_A(\log\log X)$, those with $x_1 > cx_0^2$ are $O_c(\log\log X)$, those with $x_1 > x_0\psi(x_0)$ are $o(\log X)$ whenever $\psi \to \infty$, and those with $x_1 > \lambda x_0$ are at most $\log X/\log\lambda + O(1)$ (Lemmas 8.1, 8.2, 8.3, 8.4). No arithmetic enters.

*Horizon localization.* By Kaczorowski–Pintz [9], if $\Theta > 1/2$ then at least $(\gamma_1/\pi - o(1))\log X$ distinct excursions starting below $X$ contain values of size $t^{\Theta-\eta}$. Theorem 8.11 turns every geometric count below this threshold into an equivalent form of RH: for any horizon $h$ whose crossing family has at most $(\kappa + o(1))\log X$ members with $\kappa < \gamma_1/\pi$,

$$\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \quad \text{for } x_0 < t \le \min\{x_1, h(x_0)\}.$$

In particular (Corollary 8.13)

$$\text{RH} \iff \text{Littlewood's bound holds on } x_0 < t \le \min\{x_1, \lambda x_0\}, \qquad \lambda > \lambda_* := e^{\pi/\gamma_1} = 1.2488\ldots,$$

and the square horizon $x_0^2$ is eroded from above through $cx_0^2$ and $x_0\psi(x_0)$. If RH fails, a positive proportion (on the scale $\log X$) of the Kaczorowski–Pintz excursions carry their large value in $[x_0 + \frac{1}{2}x_0^{\Theta-\eta}, \lambda x_0]$ (Corollary 8.14). The threshold $\lambda_*$ is the multiplicative distance between consecutive sign changes of $\cos(\gamma_1 \log x)$ (Remark 8.15).

*Linear window.* For $u < t \le 2u$ one has $\mu(n) = -\sum_{d|n,\, d\le u}\mu(d)$, so at a zero $u$ of $M$

$$M(t) = -\sum_{d\le u}\mu(d)\Big(\Big\lfloor\frac{t}{d}\Big\rfloor - \Big\lfloor\frac{u}{d}\Big\rfloor\Big) = -\sum_{2\le k\le t}\Big[M\Big(\frac{t}{k}\Big) - M\Big(\frac{u}{k}\Big)\Big],$$

with all arguments on the right at most $\frac{3}{4}u$ when $t \le \frac{3}{2}u$. With $\lambda = \frac{3}{2}$, RH is equivalent to the bound $t^{1/2+\varepsilon}$ for these sums on $u < t \le \min\{x_1, \frac{3}{2}u\}$ (Corollary 8.19). The recursion does not bound the window by itself: an induction on scale using only pointwise bounds loses $t^{1/2-\varepsilon}$, reduced to $t^{1/4-\varepsilon/2}$ after extracting a smooth component governed by the pole of $\zeta$ at $s = 1$ (Corollary 8.21), and any termwise contraction on the sign cone of an excursion is empty (Proposition 8.23).

*Extreme values.* If a normalized excursion statistic has a uniform tail $\Pr(Z > z) \le Ce^{-cz^p}$, the counting bounds force the maximum over a family with at most $L(X)$ members starting below $X$ to be of order $(\log L(X))^{1/p}$ in probability, with an almost sure lim sup constant $c^{-1/p}$ (Theorems 8.25 and 8.27). Each level of lacunarity removes one logarithm from the extreme scale: $\sqrt{\log X}$, $\sqrt{\log\log X}$, $\sqrt{\log\log\log X}$.

The remaining sections contain exact identities for the skeleton and its area curves (Sections 2–3), exact correlation identities and sharp bounds inside an excursion (Section 4), the relation with $\zeta$ (Section 5), area bounds and global criteria equivalent to RH (Sections 6–7), the exact bilinear form of $M$ on the square range of an excursion and the Mellin mass of an excursion (Section 8), and

the decomposition of $M$ by large prime factors, globally and inside an excursion (Sections 9–11). Section 13 records numerical data on the excursion census.

**Conventions.** $\rho = \beta + i\gamma$ runs over nontrivial zeros of $\zeta$,

$$\Theta = \sup\{\Re\rho\}, \qquad \gamma_1 = 14.134725\ldots$$

is the ordinate of the lowest zero, and RH means $\Theta = 1/2$. Littlewood's criterion [18, §14.25] states that RH holds if and only if $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$ for every $\varepsilon > 0$. Statements labelled *Heuristic* are not used in any proof.

## 2. Excursions and the regularized skeleton

Let $\mu$ be the Möbius function and

$$M(x) = \sum_{n \le x} \mu(n), \qquad x \ge 0, \tag{2.1}$$

piecewise constant and right-continuous, with jumps $\mu(n) \in \{-1, 0, 1\}$ at the integers. The segmentation of $M$ by zero runs and the record of its extrema are standard computational objects; Hurst [6] computed $M(x)$ for all $x \le 10^{16}$. The area and quadratic functionals of lattice excursions are classical [17, 14, 16, 8, 2, 1]; here they are applied to the deterministic Möbius walk.

**Definition 2.1** (Excursions and zero runs)**.** An *excursion* is an interval

$$I_\nu = (x_{0,\nu}, x_{1,\nu}]$$

with integer endpoints, $M(x_{0,\nu}) = M(x_{1,\nu}) = 0$, $M(x_{0,\nu}+1) = \sigma_\nu \in \{\pm 1\}$, and $\sigma_\nu M(n) \ge 1$ for every integer $x_{0,\nu} < n < x_{1,\nu}$. Thus $x_{1,\nu}$ is the first return to zero after the departure from $x_{0,\nu}$. Consecutive excursions may be separated by a *zero run* $[x_{1,\nu}, x_{0,\nu+1}]$, a maximal set of consecutive integers on which $M = 0$; it may consist of a single point. The first excursion is $(0, 2]$. Since $M$ takes values of both signs of size $\gg \sqrt{x}$ infinitely often (Remark 6.6) and has unit steps, it has infinitely many zeros and every excursion is finite. We also write $u_\nu = x_{0,\nu}$ and $v_\nu = x_{1,\nu}$.

Put

$$\ell_\nu = x_{1,\nu} - x_{0,\nu}, \qquad H_\nu = \max_{x_{0,\nu} < n < x_{1,\nu}} |M(n)|,$$

$$\tau_\nu = \min\{n \in I_\nu : |M(n)| = H_\nu\}, \qquad a_\nu = \tau_\nu - x_{0,\nu}, \qquad b_\nu = x_{1,\nu} - \tau_\nu.$$

Every excursion has $H_\nu \ge 1$ and $\ell_\nu \ge 2$.

**Lemma 2.2** (Height and step count)**.** *For every excursion,*

$$H_\nu \le \min(a_\nu, b_\nu) \le \frac{\ell_\nu}{2}. \tag{2.2}$$

*Proof.* The steps of $M$ are 0 or $\pm 1$, so at least $H_\nu$ steps are needed to reach height $H_\nu$ from zero and at least $H_\nu$ more to return to zero. □

**Definition 2.3** (Crossing and touching zero runs)**.** The zero run $[x_{1,\nu}, x_{0,\nu+1}]$ between consecutive excursions is a *crossing run* if $\sigma_{\nu+1} = -\sigma_\nu$ and a *touching run* if $\sigma_{\nu+1} = \sigma_\nu$.

**Definition 2.4** (Regularized skeleton)**.** The *skeleton* $M_{\mathrm{s}}$ is the continuous, piecewise linear function defined on every excursion by

$$M_{\mathrm{s}}(t) = \begin{cases} \sigma_\nu \sqrt{H_\nu}\, \dfrac{t - x_{0,\nu}}{a_\nu}, & x_{0,\nu} \le t \le \tau_\nu, \\[2mm] \sigma_\nu \sqrt{H_\nu}\, \dfrac{x_{1,\nu} - t}{b_\nu}, & \tau_\nu \le t \le x_{1,\nu}, \end{cases} \tag{2.3}$$

and by $M_{\mathrm{s}} \equiv 0$ on zero runs. The *unregularized skeleton* $M_{\mathrm{sk}}$ is defined in the same way with peak $\sigma_\nu H_\nu$.

**Definition 2.5** (Terminal block of a truncated computation)**.** Let $N \geq 2$ and $u = \max\{n \leq N : M(n) = 0\}$. If $u = N$ there is no terminal block. Otherwise $M(n) \neq 0$ for $u < n \leq N$, and since $M$ has steps $0, \pm 1$ all these values have the sign of $M(N)$. Put

$$\sigma_{\mathrm{term}} = \operatorname{sgn} M(N), \qquad H_{\mathrm{term}} = \max_{u<n\leq N} |M(n)|, \qquad \tau_{\mathrm{term}} = \min\{n \in (u, N] : |M(n)| = H_{\mathrm{term}}\}.$$

On $[u, N]$ the regularized skeleton is the segment from $(u, 0)$ to $(\tau_{\mathrm{term}}, \sigma_{\mathrm{term}}\sqrt{H_{\mathrm{term}}})$ followed, if $\tau_{\mathrm{term}} < N$, by the segment ending at $(N, \sigma_{\mathrm{term}}\sqrt{|M(N)|})$. Its signed area is

$$\Delta\mathcal{A}(u, N] = \frac{\sigma_{\mathrm{term}}}{2}\Big[\sqrt{H_{\mathrm{term}}}(N - u) + \sqrt{|M(N)|}(N - \tau_{\mathrm{term}})\Big].$$

The terminal skeleton is provisional: after the next return to zero the completed excursion acquires a different descending side. Independently of it, for real $x$ with last zero $u(x) \leq \lfloor x \rfloor$ define

$$\beta_{\mathrm{term}}(x) := \sum_{u(x)<n\leq\lfloor x\rfloor} |M(n)|, \qquad Q_{\mathrm{term}}(x) := \sum_{u(x)<n\leq\lfloor x\rfloor} M(n)^2, \qquad L_x := \lfloor x \rfloor - u(x).$$

These depend only on $M$.

**Definition 2.6** (The three curves)**.** With $M_{\mathrm{s}}^{\pm} = \max(\pm M_{\mathrm{s}}, 0)$,

$$\mathcal{A}_+(x) = \int_0^x M_{\mathrm{s}}^{+}(t)\, dt, \qquad \mathcal{A}_-(x) = -\int_0^x M_{\mathrm{s}}^{-}(t)\, dt, \qquad \mathcal{A}(x) = \int_0^x M_{\mathrm{s}}(t)\, dt. \tag{2.4}$$

$\mathcal{A}_+$ is nondecreasing, $\mathcal{A}_-$ is nonincreasing, and both are constant on zero runs and on excursions of the opposite sign.

*Remark* 2.7 (The family of $g$-skeletons). For any $g : \mathbb{N}_0 \to [0, \infty)$ with $g(0) = 0$ and $g(h) > 0$ for $h \geq 1$, one may replace the apex height $\sqrt{H}$ by $g(H)$. The signed area increment becomes $\frac{1}{2}\sigma_\nu g(H_\nu)\ell_\nu$. The correlation identities of Section 4 concern the Möbius increments and do not depend on $g$. The choice $g(H) = \sqrt{H}$ is a normalization.

## 3. Exact formulas

**Lemma 3.1** (Increment per excursion)**.** *For every complete excursion,*

$$\mathcal{A}(x_{1,\nu}) - \mathcal{A}(x_{0,\nu}) = \sigma_\nu \frac{\sqrt{H_\nu}\,\ell_\nu}{2}, \qquad \sum_{n\in I_\nu} M_{\mathrm{s}}(n) = \sigma_\nu\sqrt{H_\nu}\frac{\ell_\nu}{2}. \tag{3.1}$$

*Proof.* The two pieces are triangles of height $\sqrt{H_\nu}$ and bases $a_\nu, b_\nu$. For the discrete sum, the ascending values at $j = 1, \ldots, a_\nu$ sum to $\sqrt{H_\nu}\sum_{j\leq a_\nu} j/a_\nu = \sqrt{H_\nu}(a_\nu + 1)/2$; the descending values at $\tau_\nu + r$, $r = 1, \ldots, b_\nu$, sum to $\sqrt{H_\nu}\sum_{r\leq b_\nu}(b_\nu - r)/b_\nu = \sqrt{H_\nu}(b_\nu - 1)/2$. □

**Proposition 3.2** (Explicit form of the three curves)**.** *Let $x \in I_\nu$. Then*

$$\mathcal{A}(x) = \begin{cases} \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\dfrac{(x - x_{0,\nu})^2}{2a_\nu}, & x \leq \tau_\nu, \\[2ex] \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\left(\dfrac{a_\nu}{2} + (x - \tau_\nu) - \dfrac{(x - \tau_\nu)^2}{2b_\nu}\right), & x \geq \tau_\nu. \end{cases} \tag{3.2}$$

*At excursion endpoints,*

$$\mathcal{A}(x_{1,k}) = \sum_{\nu\leq k} \sigma_\nu\frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_+(x_{1,k}) = \sum_{\substack{\nu\leq k\\ \sigma_\nu=+1}} \frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_-(x_{1,k}) = -\sum_{\substack{\nu\leq k\\ \sigma_\nu=-1}} \frac{\sqrt{H_\nu}\ell_\nu}{2}. \tag{3.3}$$

*The curves are piecewise quadratic and $C^1$.*

*Proof.* Integrate (2.3); the endpoint formulas follow from Lemma 3.1. □

A point $c$ is a *strict inflection point* of a function if there exists $\delta > 0$ such that the function is strictly convex on one side of $c$ and strictly concave on the other, within distance $\delta$. The second derivative need not exist at $c$.

**Lemma 3.3** (Critical geometry of $\mathcal{A}$)**.** *Let $R = [p, q]$ be the zero run between an excursion of sign $\sigma_-$ ending at $p$ and the next excursion of sign $\sigma_+$ starting at $q$.*

(i) *Every apex $\tau_\nu$ is a strict inflection point.*
(ii) *If $\sigma_+ = -\sigma_-$, then $R$ is an extremal plateau; it is a strict local extremum exactly when $p = q$.*
(iii) *If $\sigma_+ = \sigma_- =: \sigma$, then $\mathcal{A}$ has no local extremum on $R$. Its concavity has sign $-\sigma$ immediately to the left and $+\sigma$ immediately to the right. If $p = q$ the point is a strict inflection; if $p < q$, no point of $R$ is a strict inflection, and the change of concavity is carried by the whole plateau.*
(iv) *Strict local extrema occur exactly at isolated crossing zeros, and plateaus of extrema exactly at crossing runs of positive length; every point of such a plateau is a non-strict local extremum.*

*Proof.* On the descending side of the left excursion, $\operatorname{sgn} \mathcal{A}' = \sigma_-$ and $\operatorname{sgn} \mathcal{A}'' = -\sigma_-$. On the ascending side of the right excursion, $\operatorname{sgn} \mathcal{A}' = \sigma_+$ and $\operatorname{sgn} \mathcal{A}'' = \sigma_+$. On $R$, $\mathcal{A}' = 0$. The assertions follow, and at every apex the sign of $\mathcal{A}''$ reverses. □

## 4. Internal correlations of an excursion

For a real block $x_1, \dots, x_L$ put

$$S_t = \sum_{j=1}^{t} x_j, \qquad C_h = \sum_{j=1}^{L-h} x_j x_{j+h}. \tag{4.1}$$

**Lemma 4.1** (Triangular identity)**.** *For every real finite sequence,*

$$\sum_{t=1}^{L-1} S_t^2 + \sum_{h=1}^{L-1} h C_h = S_L \sum_{t=1}^{L-1} S_t. \tag{4.2}$$

*Proof.* Expanding the partial sums,

$$\sum_{t=1}^{L} S_t^2 = \sum_{i,j=1}^{L} x_i x_j (L + 1 - \max(i, j)).$$

Write $\max(i, j) = \frac{1}{2}(i + j + |i - j|)$. The part containing $|i - j|$ is $\frac{1}{2} \sum_{i,j} x_i x_j |i - j| = \sum_{h=1}^{L-1} h C_h$. The remaining part is $(L + 1) S_L^2 - S_L \sum_i i x_i$, and $\sum_i i x_i = L S_L - \sum_{t=1}^{L-1} S_t$. Hence $\sum_{t=1}^{L} S_t^2 = S_L^2 + S_L \sum_{t<L} S_t - \sum_h h C_h$; subtract $S_L^2$. □

**Corollary 4.2** (Bridge case)**.** *If $S_L = 0$, then*

$$\sum_{t=1}^{L-1} S_t^2 = -\sum_{h=1}^{L-1} h C_h. \tag{4.3}$$

*Remark* 4.3 (Hermitian form). For complex $x_j$ and $C_h = \sum_{j=1}^{L-h} x_j \overline{x_{j+h}}$, the same computation gives

$$\sum_{t=1}^{L-1} |S_t|^2 + \sum_{h=1}^{L-1} h \Re C_h = \Re\left( \overline{S_L} \sum_{t=1}^{L-1} S_t \right).$$

For a complete excursion orient the increments by

$$x_j = \sigma_\nu \mu(x_{0,\nu} + j), \qquad S_t = |M(x_{0,\nu} + t)|. \tag{4.4}$$

Then $S_t > 0$ for $1 \le t < \ell_\nu$ and $S_{\ell_\nu} = 0$. Put

$$Q_\nu = \sum_{t=1}^{\ell_\nu - 1} S_t^2, \qquad C_h^{(\nu)} = \sum_{j=1}^{\ell_\nu - h} x_j x_{j+h}, \qquad \text{so that} \qquad Q_\nu = -\sum_{h=1}^{\ell_\nu - 1} h C_h^{(\nu)}. \tag{4.5}$$

**Lemma 4.4** (Zeroth moment and extreme lag)**.** *With $q_\nu = \sum_j x_j^2$,*

$$\sum_{h=1}^{\ell_\nu - 1} C_h^{(\nu)} = -\frac{q_\nu}{2}, \qquad C_{\ell_\nu - 1}^{(\nu)} = -1, \tag{4.6}$$

*and $q_\nu$ is even and at least* 2*.*

*Proof.* $0 = (\sum_j x_j)^2 = q_\nu + 2\sum_h C_h^{(\nu)}$. The first oriented step is $+1$; immediately before the first return the path is at height 1, so the last step is $-1$. The bridge condition forces equally many positive and negative active steps. □

**Proposition 4.5** (Quadratic decomposition by excursions)**.** *For real $x \ge 1$ let $u(x)$ and $L_x$ be as in Definition 2.5, and for $1 \le j \le L_x$ put $y_j = \mu(u(x) + j)$ and $C_h^{\mathrm{term}} = \sum_{j=1}^{L_x - h} y_j y_{j+h}$. Then*

$$\sum_{n \le x} M(n)^2 = \sum_{\nu : x_{1,\nu} \le u(x)} Q_\nu + Q_{\mathrm{term}}(x), \tag{4.7}$$

$$Q_{\mathrm{term}}(x) = M(\lfloor x \rfloor)^2 + M(\lfloor x \rfloor) \sum_{t=1}^{L_x - 1} M(u(x) + t) - \sum_{h=1}^{L_x - 1} h C_h^{\mathrm{term}}, \tag{4.8}$$

*with $Q_{\mathrm{term}}(x) = 0$ if $L_x = 0$. At a zero $x$ of $M$,*

$$\sum_{n \le x} M(n)^2 = \sum_{x_{1,\nu} \le x} Q_\nu = -\sum_{x_{1,\nu} \le x} \sum_h h C_h^{(\nu)}. \tag{4.9}$$

*Proof.* Every integer up to $u(x)$ lies either in a zero run or in a unique excursion interior. Apply Lemma 4.1 to the terminal block and add its final term $S_{L_x}^2$. □

Thus the quadratic sum sees only lag correlations internal to excursions: pairs of Möbius increments in different completed excursions do not contribute.

Let

$$\beta_\nu = \sum_{t=1}^{\ell_\nu - 1} S_t = \sum_{n \in I_\nu} |M(n)|. \tag{4.10}$$

Cauchy–Schwarz gives

$$\beta_\nu^2 \le (\ell_\nu - 1) Q_\nu, \tag{4.11}$$

with equality exactly for a flat excursion of height 1. To separate support from sign geometry, let $P_h^{(\nu)} = \#\{j : x_j x_{j+h} \ne 0\}$ and let $D_h^{(\nu)}$ count the opposite-sign pairs among them. Then

$$C_h^{(\nu)} = P_h^{(\nu)} - 2D_h^{(\nu)}, \qquad Q_\nu = \sum_{h=1}^{\ell_\nu - 1} h\big(2D_h^{(\nu)} - P_h^{(\nu)}\big). \tag{4.12}$$

**Proposition 4.6** (Sharp bounds at fixed length and height)**.** *Let $S_0 = S_L = 0$, $S_t \geq 1$ for $0 < t < L$, $S_t - S_{t-1} \in \{-1, 0, 1\}$ and $H = \max S_t$. Then $L \geq 2H$ and*

$$H^2 + L - 2H \leq \beta \leq H(L - H), \tag{4.13}$$

$$\frac{2H^3 + H}{3} + L - 2H \leq Q \leq \frac{2H^3 + H}{3} + (L - 2H)H^2. \tag{4.14}$$

*All four bounds are attained. In particular*

$$Q \geq \frac{2H^3 + H}{3}, \qquad Q \geq L - 1. \tag{4.15}$$

*Proof.* Before and after the first visit to $H$ the path visits each level $1, \ldots, H-1$. Together with one visit to $H$, these $2H - 1$ designated interior times contribute $2(1 + \cdots + (H-1)) + H = H^2$ to $\beta$ and $2\sum_{j<H} j^2 + H^2 = (2H^3 + H)/3$ to $Q$. The remaining $L - 2H$ interior times lie at heights between 1 and $H$; placing them all at height 1, or all at height $H$, attains the bounds. □

**Corollary 4.7** (Sharp fixed-length bounds)**.** *For $L \geq 2$,*

$$\beta \leq \left\lfloor \frac{L^2}{4} \right\rfloor, \qquad Q \leq \begin{cases} L(L^2+2)/12, & L \text{ even}, \\ L(L^2-1)/12, & L \text{ odd}, \end{cases}$$

*and in particular*

$$Q \leq \frac{L^3 + 2L}{12}. \tag{4.16}$$

*Both bounds are attained by the tent path, with one horizontal step at maximal height when $L$ is odd.*

*Proof.* Use $S_t \leq \min(t, L - t)$ and sum this bound and its square. □

## 5. Where $\zeta$ enters

The black curve depends on $M$ nonlinearly, through $\sqrt{H_\nu}$ and through the selection of $x_{0,\nu}, \tau_\nu, x_{1,\nu}$, and no closed expression of its Mellin transform in terms of $\zeta$ is known. The nodes $x_{0,\nu}, x_{1,\nu}$ are zeros of $M$. If RH holds and all zeros are simple, there is a sequence $T_k \to \infty$ such that, for non-integer $x$,

$$M(x) = \lim_{k\to\infty} \sum_{|\gamma|<T_k} \frac{x^\rho}{\rho\,\zeta'(\rho)} - 2 + \sum_{n\geq 1} \frac{(-1)^{n-1}(2\pi/x)^{2n}}{(2n)!\,n\,\zeta(2n+1)} \tag{5.1}$$

[18, §14.27]. For $\widetilde{A}(x) := \int_0^x M(t)\,dt$ one has, for $\Re s > 1$,

$$\int_1^\infty \widetilde{A}(x) x^{-s-2}\,dx = \frac{1}{s(s+1)\zeta(s)}. \tag{5.2}$$

Shifting the contour formally, the residues are

$$\sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)\zeta'(\rho)} - 2x + 12 + (\text{residues at the trivial zeros}), \tag{5.3}$$

where $x/\zeta(0) = -2x$ is the residue at $s = 0$ and $1/((-1)\zeta(-1)) = 12$ the residue at $s = -1$. The factor $(\rho+1)^{-1}$ improves convergence, but RH and simplicity alone do not control $1/\zeta'(\rho)$ well enough to turn (5.3) into an identity with an untruncated sum; results of this type require additional input of Gonek–Hejhal type [13]. The area of $M_{\mathrm{sk}}$ differs from $\widetilde{A}$ by explicit local terms.

**Proposition 5.1** (Dipole identity)**.** *For each excursion let its* legs *be the integer sets* $\lambda = (u_\nu, \tau_\nu] \cap \mathbb{Z}$ *and* $(\tau_\nu, v_\nu] \cap \mathbb{Z}$*, with midpoints* $c_\lambda = (u_\nu + \tau_\nu)/2$ *and* $(\tau_\nu + v_\nu)/2$*. Then for every zero* $v_k$*,*

$$\int_0^{v_k} M_{\mathrm{sk}}(t)\,dt = \widetilde{A}(v_k) + \sum_{\lambda \subset (0, v_k]} \sum_{m \in \lambda} \mu(m)\,(m - c_\lambda). \tag{5.4}$$

*Proof.* Since $M$ is constant on each $[n, n+1)$, $\int_a^b M(t)\,dt = \sum_{n=a}^{b-1} M(n)$ for integers $a < b$. On an ascent leg, $\int_u^\tau M_{\mathrm{sk}} = \sigma H a/2$ and $\int_u^\tau M = \sum_{m=u+1}^{\tau} \mu(m)(\tau - m)$, while $\sigma H = \sum_{m=u+1}^{\tau} \mu(m)$; the difference is $\sum_m \mu(m)(a/2 - \tau + m) = \sum_m \mu(m)(m - c_\lambda)$. On a descent leg, $\int_\tau^v M_{\mathrm{sk}} = \sigma H b/2$ and $\int_\tau^v M = b\,\sigma H + \sum_{m=\tau+1}^{v} \mu(m)(v - m)$ with $\sum_{m=\tau+1}^{v} \mu(m) = -\sigma H$; the difference is $\sum_m \mu(m)(m - (v - b/2))$, and $v - b/2 = (\tau + v)/2 = c_\lambda$. On zero runs both integrands vanish. Sum over legs. □

## 6. Bounds for the area curves

**Lemma 6.1** (Transfer)**.** *Let* $x \geq 1$ *and let* $x^* \geq x$ *be a zero of* $M$ *(in a computation truncated at* $N \geq x$*, take* $x^* = N$ *and the terminal skeleton of Definition 2.5). If* $|M(n)| \leq c\,n^{1/2}$ *for all* $1 \leq n \leq x^*$ *(so necessarily* $c \geq 1$*), then*

$$|M_{\mathrm{s}}(t)| \leq c^{1/2} t^{1/4} \quad (0 \leq t \leq x), \qquad \mathcal{A}_+(x) + |\mathcal{A}_-(x)| \leq \tfrac{4}{5}\, c^{1/2} x^{5/4}. \tag{6.1}$$

*Proof.* For $t \geq \tau_\nu$, $|M_{\mathrm{s}}(t)| \leq \sqrt{H_\nu} \leq c^{1/2}\tau_\nu^{1/4} \leq c^{1/2} t^{1/4}$. For $u_\nu \leq t \leq \tau_\nu$ the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$ lies below the chord of $c^{1/2}t^{1/4}$ over $[u_\nu, \tau_\nu]$, whose endpoints $c^{1/2}u_\nu^{1/4} \geq 0$ and $c^{1/2}\tau_\nu^{1/4} \geq \sqrt{H_\nu}$ (as $\tau_\nu \leq x^*$) are at least as high; that chord lies below $c^{1/2}t^{1/4}$ by concavity. Integrate. □

**Lemma 6.2** (A concave majorant)**.** *Let* $\phi(y) = (\log y)^{3/5}(\log\log y)^{-1/5}$ *and, for* $\kappa > 0$*,* $F(y) = y^{1/2} e^{-\kappa\phi(y)}$*. There is* $y_0 = y_0(\kappa)$ *such that* $F$ *is increasing and concave on* $[y_0, \infty)$*.*

*Proof.* With $y = e^s$ and $g(s) = s^{3/5}(\log s)^{-1/5}$, one has $F = \exp(s/2 - \kappa g(s))$. Put $h(s) = \frac{1}{2} - \kappa g'(s)$. Then

$$yF'(y) = F\,h(s), \qquad y^2 F''(y) = F\big(h(s)^2 - h(s) - \kappa g''(s)\big).$$

Since $g'(s) \to 0$ and $g''(s) \to 0$, $h(s) \to \frac{1}{2}$ and $h^2 - h - \kappa g'' \to -\frac{1}{4}$. Hence $F' > 0$ and $F'' < 0$ for large $y$. □

**Corollary 6.3.** *From the Vinogradov–Korobov bound* [19]

$$M(x) \ll x \exp(-c\,\phi(x)) \tag{6.2}$$

*one obtains*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{3/2} \exp(-\tfrac{c}{2}\,\phi(x)). \tag{6.3}$$

*Proof.* Let $K$ be such that $|M(y)| \leq K y\, e^{-c\phi(y)}$ for $y \geq y_1$, and let $F$ be as in Lemma 6.2 with $\kappa = c/2$, so that $\sqrt{|M(y)|} \leq K^{1/2} F(y)$ for $y \geq y_1$. Only finitely many excursions start below $Y_0 := \max(y_0, y_1)$; they are finite, so they contribute $O(1)$ to the area. Consider an excursion with $u_\nu \geq Y_0$. Its apex satisfies $\sqrt{H_\nu} = \sqrt{|M(\tau_\nu)|} \leq K^{1/2} F(\tau_\nu)$; the bound is taken at the apex $\tau_\nu$, which lies in the excursion, and not at its future endpoint $v_\nu$. For $t \geq \tau_\nu$ monotonicity of $F$ gives $|M_{\mathrm{s}}(t)| \leq \sqrt{H_\nu} \leq K^{1/2} F(t)$. For $u_\nu \leq t \leq \tau_\nu$, $|M_{\mathrm{s}}(t)|$ is the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$. This chord lies below the chord from $(u_\nu, K^{1/2}F(u_\nu))$ to $(\tau_\nu, K^{1/2}F(\tau_\nu))$, which has both endpoints at least as high, and the latter lies below $K^{1/2}F$ on $[u_\nu, \tau_\nu] \subset [y_0, \infty)$ by concavity. Hence $|M_{\mathrm{s}}(t)| \leq K^{1/2} F(t)$ for all $t \geq Y_0$ outside the finitely many early excursions, and since $F$ is increasing,

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \leq O(1) + K^{1/2} \int_{Y_0}^{x} F(t)\,dt \leq O(1) + K^{1/2} x F(x) = O(1) + K^{1/2} x^{3/2} e^{-\frac{c}{2}\phi(x)}. \quad \square$$

**Corollary 6.4.** *If RH holds, then*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll_\varepsilon x^{5/4+\varepsilon}. \tag{6.4}$$

*Consequently, if $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \gg x^{5/4+\delta}$ along an unbounded sequence for some fixed $\delta > 0$, then RH is false.*

*Remark* 6.5 (Endpoint-indexed bounds). From $\ell_\nu \geq 2H_\nu$ and $\sqrt{H_\nu}\ell_\nu \ll v_\nu^{5/4+\varepsilon}$ one deduces only $H_\nu \ll v_\nu^{5/6+O(\varepsilon)}$. This bound is indexed by the *future* endpoint $v_\nu$ of the excursion; it gives no bound for $M(x)$ in terms of $x$ at interior points, and hence no zero-free half-plane. Whether (6.4) is equivalent to RH is open.

*Remark* 6.6 (Oscillation of the heights). The Mertens conjecture is false [15, 10]; the current bounds are due to Hurst [6], who proves $\limsup M(x)/\sqrt{x} > 1.826054$ and $\liminf M(x)/\sqrt{x} < -1.837625$. The excursion containing a point $n$ with $M(n) > 1.826054\sqrt{n}$ has $u_\nu < n$, so $H_\nu > 1.826054\, u_\nu^{1/2}$ for infinitely many positive excursions, and similarly with 1.837625 for negative ones. A lower bound indexed by $v_\nu$ does not follow, since $v_\nu$ may be much larger than $n$.

## 7. Global criteria equivalent to RH

The Mellin identity and the global $L^1/L^2$ equivalences below are consequences of Littlewood's criterion and partial summation; the excursion decompositions and pruning lemmas repackage them in terms of excursions. For $\Re s > 1$,

$$\int_1^\infty M(x)x^{-s-1}\,dx = \frac{1}{s\,\zeta(s)}. \tag{7.1}$$

If $\int_1^\infty |M(x)|x^{-\sigma-1}dx < \infty$ for every $\sigma > 1/2$, the left side continues analytically to $\Re s > 1/2$, so $\zeta$ has no zeros there.

**Proposition 7.1** ($L^1$ criterion)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{v_\nu \leq x} \beta_\nu + \beta_{\mathrm{term}}(x) = \sum_{n \leq x} |M(n)| \ll x^{3/2+\varepsilon}. \tag{7.2}$$

*Proof.* RH gives $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$. Conversely, for $\sigma > 1/2$ choose $0 < \varepsilon < \sigma - 1/2$; partial summation gives $\sum_n |M(n)|n^{-\sigma-1} < \infty$, and since $M$ is constant on $[n, n+1)$ the remark after (7.1) applies. □

**Lemma 7.2** (Pruning)**.** $\beta_\nu \leq H_\nu(\ell_\nu - 1) \leq \ell_\nu^2/2$, *hence*

$$\sum_{v_\nu \leq x,\ \ell_\nu \leq L_0} \beta_\nu \leq \frac{xL_0}{2}. \tag{7.3}$$

**Corollary 7.3** (Reduction to long excursions)**.** *For fixed $\delta \geq 0$ and $L_0 = x^{1/2-\delta}$, RH holds if and only if*

$$\sum_{\substack{v_\nu \leq x \\ \ell_\nu > L_0}} \beta_\nu + \mathbf{1}_{\{L_x > L_0\}}\beta_{\mathrm{term}}(x) \ll x^{3/2+\varepsilon} \quad \text{for every } \varepsilon > 0. \tag{7.4}$$

*At most $x/L_0$ long excursions are involved.*

*Remark* 7.4 (The $\sqrt{x}$ frontier). The cut may depend on $\varepsilon$: with $L_0 = x^{1/2+\varepsilon/2}$ the discarded part is $\leq x^{3/2+\varepsilon/2}/2$. The frontier of pruning is therefore $x^{1/2+o(1)}$; what requires new information is a cut $x^\theta$ with a fixed $\theta > 1/2$. For $\ell_\nu \asymp x$ the trivial bound gives $\beta_\nu \ll x^2$ against a budget $x^{3/2+\varepsilon}$, and there are only $O(1)$ such excursions.

**Heuristic 7.5** (Renewal tail)**.** If excursion lengths behave like those of a recurrent random walk,

$$\#\{v_\nu \le x : \ \ell_\nu > L\} \asymp \sqrt{x}\, L^{-1/2}, \tag{7.5}$$

and with $\beta_\nu \asymp \ell_\nu^{3/2}$ the dyadic range $[L, 2L]$ contributes $\asymp \sqrt{x}\, L$ to (7.2), so the terminal scales $L \asymp x$ dominate the $L^1$ sum.

**Theorem 7.6** (Correlation criterion)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{n\le x} M(n)^2 = \sum_{v_\nu \le x} Q_\nu + Q_{\mathrm{term}}(x) = -\sum_{v_\nu \le x} \sum_h h C_h^{(\nu)} + Q_{\mathrm{term}}(x) \ll x^{2+\varepsilon}. \tag{7.6}$$

*Proof.* Under RH, $M(n)^2 \ll_\varepsilon n^{1+\varepsilon}$. Conversely, by Cauchy–Schwarz,

$$\sum_{n\le x} |M(n)| \le x^{1/2}\Big(\sum_{n\le x} M(n)^2\Big)^{1/2} \ll x^{3/2+\varepsilon/2};$$

apply Proposition 7.1. □

**Corollary 7.7** (Local sufficient conditions)**.** *Suppose that for every $\varepsilon > 0$ there is $C_\varepsilon$ such that*

$$Q_\nu \le C_\varepsilon\, \ell_\nu^{2+\varepsilon} \quad \text{for every complete excursion with } \ell_\nu > v_\nu^{1/2}, \tag{7.7}$$

*and $Q_{\mathrm{term}}(x) \le C_\varepsilon L_x^{2+\varepsilon}$ whenever $L_x > x^{1/2}$. Then RH holds. In particular it suffices that* (7.7) *hold for every complete excursion.*

*Proof.* Split the completed excursions at $\ell_\nu = x^{1/2}$. If $\ell_\nu \le x^{1/2}$, then (4.16) gives $Q_\nu \le \ell_\nu x/12 + \ell_\nu/6$, and disjointness bounds the total by $x^2/12 + x/6$. If $\ell_\nu > x^{1/2}$ then $\ell_\nu > v_\nu^{1/2}$ since $v_\nu \le x$, and the hypothesis bounds the total by $C_\varepsilon \sum \ell_\nu^{2+\varepsilon} \le C_\varepsilon (\sum \ell_\nu)^{2+\varepsilon} \le C_\varepsilon x^{2+\varepsilon}$. The terminal block is not a complete excursion, so (7.7) does not apply to it; it is bounded by (4.16) if $L_x \le x^{1/2}$ and by the terminal hypothesis otherwise. Apply Theorem 7.6. □

**Heuristic 7.8** (Ballistic excursions)**.** In a model where, after a zero of $M$, the next values of $\mu$ behave locally at random with squarefree density $6/\pi^2$, the pattern "$k$ non-zero steps of one sign followed by $k$ of the other" follows a given zero with probability $\ge c_0^{-k}$. If excursions with $H_\nu = k$ and $\ell_\nu \asymp k$ occur for unbounded $k$, then by (4.15) $Q_\nu/\ell_\nu^{2+\varepsilon} \gg k^{1-\varepsilon} \to \infty$, so the unpruned version of (7.7) fails for every $\varepsilon \in (0,1)$. Such excursions have $\ell_\nu \ll v_\nu^{1/2}$ and do not affect the pruned condition (7.7); this is why the pruning in Corollary 7.7 is needed.

**Theorem 7.9** (Energy criterion)**.** *With $u$ the last zero $\le x$ and $H_{\mathrm{run}}(x) = \max_{u<n\le x} |M(n)|$, define*

$$\mathcal{E}(x) := \frac{1}{4}\Big(\sum_{v_\nu \le x} H_\nu \ell_\nu + (x-u) H_{\mathrm{run}}(x)\Big). \tag{7.8}$$

*For a complete excursion let $J$ be the chord joining $(u_\nu, \mathcal{A}(u_\nu))$ to $(v_\nu, \mathcal{A}(v_\nu))$. By* (3.1)*, $J' = \sigma_\nu \sqrt{H_\nu}/2$, so $\frac{1}{4} H_\nu \ell_\nu = \int_{I_\nu} (J')^2$ is the Dirichlet energy of the chord. Then*

$$\sum_{n\le x} |M(n)| \le 4\mathcal{E}(x) \le x \max_{n\le x} |M(n)|, \tag{7.9}$$

*and RH holds if and only if $\mathcal{E}(x) \ll x^{3/2+\varepsilon}$ for every $\varepsilon > 0$.*

*Proof.* $\beta_\nu \le H_\nu \ell_\nu$, $\beta_{\mathrm{term}}(x) \le (x-u) H_{\mathrm{run}}(x)$ and $\sum \ell_\nu + (x-u) \le x$; apply Proposition 7.1. □

## 8. Lacunary excursion families and horizon-localized criteria

8.1. **Geometric counts.** The lemmas of this subsection use only that excursions are disjoint integer intervals ordered by their starting points, the next one starting at or after the end of the previous one. They therefore hold verbatim for every sequence of disjoint integer intervals, deterministic or random, and in particular pathwise for any stochastic model of excursions (Section 8.5). For a family $\mathcal{F}$ of excursions and $X \geq 2$ put

$$E_{\mathcal{F}}(X) := \#\{\nu \in \mathcal{F} : x_{0,\nu} \leq X\}.$$

For a function $h$ on $[0,\infty)$ with $h(x) \geq x$, the family of excursions *crossing the horizon* $h$ is

$$\mathcal{F}_h := \{\nu : x_{1,\nu} > h(x_{0,\nu})\}.$$

For $A > 1$ the excursion $I_\nu$ is *$A$-exceptional* if $x_{1,\nu} > x_{0,\nu}^A$, i.e. $\nu \in \mathcal{F}_{x^A}$, and *$A$-ordinary* otherwise; for $A = 2$ we say *square-exceptional* and *square-ordinary*. For $\lambda > 1$ it is *$\lambda$-long* if $x_{1,\nu} > \lambda x_{0,\nu}$. We write $E_A = E_{\mathcal{F}_{x^A}}$, $E_\square = E_2$ and $E_\lambda = E_{\mathcal{F}_{\lambda x}}$; the context separates $E_A$ ($A$ an exponent) from $E_\lambda$ ($\lambda$ a factor).

**Lemma 8.1** (Power lacunarity)**.** *For every $A > 1$ and $X \geq 2$,*

$$E_A(X) \leq 2 + \frac{1}{\log A} \log \frac{\log X}{\log 2}. \tag{8.1}$$

*In particular*

$$E_\square(X) \leq 2 + \log_2 \frac{\log X}{\log 2}, \qquad E_\square(X) \leq 4 \log\log X \quad (X \geq e^e). \tag{8.2}$$

*Proof.* No excursion starts at 1, since $M(1) = 1$. List the $A$-exceptional starts $\geq 2$ as $2 \leq a_1 < a_2 < \cdots$. The next excursion starts at or after the end of the preceding one, so $a_{j+1} \geq x_{1,j} > a_j^A$ and $\log a_j \geq A^{j-1} \log 2$. If $a_j \leq X$ then $j \leq 1 + \log(\log X/\log 2)/\log A$; add 1 for the excursion $(0,2]$. For the second bound in (8.2), $\log_2(\log X/\log 2) = (\log\log X - \log\log 2)/\log 2 \leq 1.443 \log\log X + 0.529$. □

The next lemma is the partition of the square horizon $[x_0, x_0^2] = [x_0, cx_0^2] \cup [cx_0^2, x_0^2]$: the upper part is accessible only to a doubly lacunary family, for every fixed $c > 0$.

**Lemma 8.2** (Fractional square horizon)**.** *For $0 < c \leq 1$ and $X \geq 2$,*

$$E_{\mathcal{F}_{cx^2}}(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > cx_{0,\nu}^2\} \leq \frac{2}{c} + 2 + \log_2 \frac{\log X}{\log 2}. \tag{8.3}$$

*Consequently, the range $cx_0^2 < t \leq x_0^2$ of the excursions starting at most $X$ is visited by at most $O_c(\log\log X)$ excursions.*

*Proof.* At most $\lceil 2/c \rceil \leq 2/c + 1$ members start below $2/c$. List the others as $a_1 < a_2 < \cdots$ and put $b_j = ca_j \geq 2$. Then $a_{j+1} \geq x_{1,j} > ca_j^2$, so

$$b_{j+1} = ca_{j+1} > c^2 a_j^2 = b_j^2,$$

and $\log b_j \geq 2^{j-1} \log 2$. Since $b_j \leq cX \leq X$, the number of such $j$ is at most $1 + \log_2(\log X/\log 2)$. An excursion containing a point $t > cx_0^2$ has $x_1 \geq t > cx_0^2$. □

**Lemma 8.3** (Superlinear horizons)**.** *Let $\psi : [1,\infty) \to (0,\infty)$ satisfy $\psi(x) \to \infty$. Then*

$$E_{\mathcal{F}_{x\psi(x)}}(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > x_{0,\nu}\psi(x_{0,\nu})\} = o(\log X).$$

*Proof.* Given $K > 0$ choose $x_K$ with $\psi(x) \geq e^K$ for $x \geq x_K$. At most $x_K + 1$ members start below $x_K$. For consecutive members $a_j < a_{j+1}$ starting at or above $x_K$, $a_{j+1} \geq x_{1,j} > a_j\psi(a_j) \geq e^K a_j$, so $\log a_j$ increases by more than $K$ at each step and there are at most $1 + \log X/K$ of them. Hence $\limsup E(X)/\log X \leq 1/K$ for every $K$. □

**Lemma 8.4** (Linear lacunarity)**.** *For every $\lambda > 1$ and $X \geq 2$,*

$$E_\lambda(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > \lambda x_{0,\nu}\} \leq 2 + \frac{\log(X/2)}{\log \lambda}. \tag{8.4}$$

*Proof.* For starts $a_j \geq 2$, $a_{j+1} \geq x_{1,j} > \lambda a_j$, hence $a_j > 2\lambda^{j-1}$ for $j \geq 2$; add 1 for $(0, 2]$. □

All four bounds are sharp up to constants for general families of disjoint intervals, as geometric and doubly geometric chains of intervals show.

8.2. **Arithmetic input: Kaczorowski–Pintz.** We separate the published statement, its translation from $M_0 = M + 2$ to $M$, and the consequences for excursions.

**Theorem 8.5** (Kaczorowski–Pintz [9])**.** *Let $M_0(x) = M(x) - 1/\zeta(0) = M(x) + 2$, the function $M_a$ of* [9] *at $a = 0$. Let $V(f, Y)$ be the number of sign changes of $f$ on $[1, Y]$ counted through alternating-sign points, as defined in* [9]*. Then:*

(i) [9, Corollary 4, p. 178, (2.13)–(2.15)]

$$\liminf_{Y\to\infty} \frac{V(M_0, Y)}{\log Y} \geq \frac{\gamma_1}{\pi}, \tag{8.5}$$

*with combined oscillation of size $x^{\Theta-\eta}$ for every $\eta > 0$, in the sense of their definition following (1.1): the alternating points $t_j$ counted by $V$ can be chosen with $|M_0(t_j)| \geq t_j^{\Theta-\eta}$;*

(ii) *Corollary 4 is deduced from their Corollary 1, which asserts relations (2.1)–(2.3) of their Theorem 1* [9, pp. 175–176]*; in particular, for every $\delta > 0$ and $Y > Y_0(\delta)$, the interval $[Y^{1-\delta}, Y]$ contains two points $t_+, t_-$ with $M_0(t_+) \geq t_+^{\Theta-\eta}$ and $M_0(t_-) \leq -t_-^{\Theta-\eta}$.*

**Lemma 8.6** (From $M_0$ to $M$)**.** *If $|M_0(t)| > 2$, then $\operatorname{sgn} M(t) = \operatorname{sgn} M_0(t)$ and $|M(t)| \geq |M_0(t)| - 2$. In particular, for points with $|M_0(t)| \geq t^{\Theta-\eta}$ and $t$ large, $\operatorname{sgn} M(t) = \operatorname{sgn} M_0(t)$ and $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$.*

*Proof.* $M_0 > 2$ gives $M > 0$; $M_0 < -2$ gives $M < -4$. The second assertion uses $t^{\Theta-\eta} \to \infty$ for $\eta < \Theta$. □

**Corollary 8.7** (Distinct large excursions)**.** *If $\Theta > 1/2$ and $0 < \eta < \Theta - 1/2$, then*

$$N_\eta(Y) \geq \left(\frac{\gamma_1}{\pi} - o(1)\right) \log Y, \tag{8.6}$$

*where $N_\eta(Y)$ counts distinct excursions starting at most $Y$ that contain a point $t \leq Y$ with $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$.*

*Proof.* Take the alternating points of Theorem 8.5(i) in $[1, Y]$, discarding the boundedly many below the threshold of Lemma 8.6. By that lemma, consecutive points $t < t'$ have $M(t)$ and $M(t')$ of opposite signs. Since $M$ is integer valued with steps $0, \pm 1$, it vanishes at some integer in $(t, t')$, so $t$ and $t'$ lie in different excursions. Each point with $M(t) \neq 0$ belongs to exactly one excursion, and excursions are ordered, so the excursions containing the ordered points are pairwise distinct. Their starts are below the points, hence at most $Y$, and $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$ at each point. □

**Corollary 8.8** (Multiplicative localization of Mertens returns)**.** *For every fixed $A > 1$ there exists $U_A$ such that every excursion starting at $u \geq U_A$ ends at $v \leq u^A$. Equivalently, $M$ has only finitely many $A$-exceptional excursions; in particular only finitely many square-exceptional ones.*

*Proof.* Put $\delta = 1 - 1/A$ and $Y = u^A$, so that $[Y^{1-\delta}, Y] = [u, u^A]$, and fix $0 < \eta < \Theta$ (possible since $\Theta \geq 1/2$). For $u$ large, Theorem 8.5(ii) and Lemma 8.6 give $t_+, t_- \in [u, u^A]$ with $M(t_+) > 0 > M(t_-)$. Neither point lies in $[u, u + 1)$, because $M(u) = 0$; so $u < \lfloor t_\pm \rfloor$. Since $M$ changes by $0, \pm 1$ at each integer, it vanishes at some integer $n$ strictly between $\lfloor t_+ \rfloor$ and $\lfloor t_- \rfloor$, and $u < n < u^A$. If $u = x_{0,\nu}$, the first return satisfies $x_{1,\nu} \leq n < u^A$. □

*Remark* 8.9. A bare sign change of $M_0 = M + 2$ would not force a zero of $M$. The decisive point is that the local sign change is combined with oscillation of size $t^{\Theta-\eta} \to \infty$, so that Lemma 8.6 applies to both witnesses.

*Remark* 8.10 (Geometric counts and arithmetic localization). For the actual Mertens excursions Corollary 8.8 is stronger than Lemma 8.1: each $A$-exceptional family is finite. The geometric lemmas remain separate results because they are elementary, independent of $\zeta$, and valid for every ordered family of disjoint intervals. This is exactly what the localization theorem below and the extreme-value theorems of Section 8.5 use: the former needs only a count below $(\gamma_1/\pi)\log X$, the latter needs a count valid pathwise in a model where no analogue of Kaczorowski–Pintz is available.

### 8.3. **Horizon localization.**

**Theorem 8.11** (Horizon-localized RH criterion)**.** *Let $h$ be a function on $[0,\infty)$ with $h(x) \geq x$, let $\mathcal{F}$ be a family of excursions, and suppose that for some $\kappa < \gamma_1/\pi$*

$$E_{\mathcal{F}\cup\mathcal{F}_h}(X) \leq (\kappa + o(1))\log X \qquad (X \to \infty). \tag{8.7}$$

*Then RH is equivalent to the following statement: for every $\varepsilon > 0$ there is $C_\varepsilon$ such that*

$$|M(t)| \leq C_\varepsilon t^{1/2+\varepsilon} \qquad \text{for all } t \in I_\nu,\ \nu \notin \mathcal{F},\ t \leq h(x_{0,\nu}). \tag{8.8}$$

*No condition is imposed on the excursions in $\mathcal{F}$, nor on the part of any excursion lying beyond its horizon $h(x_0)$.*

*Proof.* RH implies (8.8) by Littlewood's criterion. Conversely, suppose $\Theta > 1/2$ and fix $\eta, \varepsilon > 0$ with $1/2 + \varepsilon < \Theta - \eta$. By (8.6) there are at least $(\gamma_1/\pi - o(1))\log X$ distinct excursions starting at most $X$, each containing a witness $t$ with $|M(t)| \geq \frac{1}{2}t^{\Theta-\eta}$. Consider such an excursion $\nu$ with witness $t$. If $\nu \notin \mathcal{F}$ and $t \leq h(x_{0,\nu})$, the witness lies in the range of (8.8). Otherwise $\nu \in \mathcal{F}$, or $t > h(x_{0,\nu})$, in which case $x_{1,\nu} \geq t > h(x_{0,\nu})$ and $\nu \in \mathcal{F}_h$. By (8.7) at least

$$\Big(\frac{\gamma_1}{\pi} - \kappa - o(1)\Big)\log X \longrightarrow \infty$$

witnesses lie in the range of (8.8). They belong to distinct excursions, so they are unbounded, and at them $t^{\Theta-\eta}/t^{1/2+\varepsilon} \to \infty$, contradicting (8.8). □

The case $h(x) = \infty$ excludes a sparse family $\mathcal{F}$ of whole excursions; the case $\mathcal{F} = \emptyset$ truncates every excursion at its horizon.

**Corollary 8.12** (Square horizon)**.** *For every fixed $A > 1$, RH is equivalent to* (8.8) *restricted to $A$-ordinary excursions. In particular*

$$\boxed{\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \text{ for all } t \text{ in excursions with } x_1 \leq x_0^2,} \tag{8.9}$$

*and, equivalently, RH holds if and only if $|M(t)| \ll_\varepsilon t^{1/2+\varepsilon}$ for $x_{0,\nu} < t \leq \min\{x_{1,\nu}, x_{0,\nu}^2\}$ over all excursions.*

*Proof.* Take $\mathcal{F} = \mathcal{F}_{x^A}$, $h = \infty$, respectively $\mathcal{F} = \emptyset$, $h(x) = x^2$, in Theorem 8.11; Lemma 8.1 gives $\kappa = 0$. □

The count $E_\square(X) = O(\log\log X)$ is thus what makes the square horizon admissible. The square horizon can be lowered.

**Corollary 8.13** (Descending horizons)**.** *RH is equivalent to Littlewood's bound $|M(t)| \ll_\varepsilon t^{1/2+\varepsilon}$ on the ranges*

$$x_{0,\nu} < t \leq \min\{x_{1,\nu}, h(x_{0,\nu})\}$$

*over all excursions, for each of the following horizons:*

(i) *$h(x) = cx^2$, for any fixed $c > 0$;*

(ii) $h(x) = x\psi(x)$, *for any* $\psi$ *with* $\psi(x) \to \infty$;
(iii) $h(x) = \lambda x$, *for any fixed* $\lambda > \lambda_* := e^{\pi/\gamma_1} = 1.24889\ldots$

*The same holds with "truncate at* $h(x_0)$*" replaced by "discard the excursions with* $x_1 > h(x_0)$*". In particular it is enough to control the excursions with* $x_1 \le \frac{5}{4}x_0$.

*Proof.* Apply Theorem 8.11 with $\mathcal{F} = \emptyset$ (respectively $\mathcal{F} = \mathcal{F}_h$, $h = \infty$). For (i), $c \ge 1$ reduces to Lemma 8.1 and $0 < c < 1$ to Lemma 8.2; $\kappa = 0$. For (ii), Lemma 8.3 gives $\kappa = 0$. For (iii), Lemma 8.4 gives $\kappa = 1/\log\lambda$, and $1/\log\lambda < \gamma_1/\pi$ exactly when $\lambda > \lambda_*$. Finally $5/4 > \lambda_*$. □

Case (iii) implies (i) and (ii): in both, $h(x_0) \ge \lambda x_0$ for large $x_0$, so the bound on the larger ranges implies it on the ranges of (iii), the finitely many remaining excursions affecting only $C_\varepsilon$. The horizon has been lowered from $x_0^2$ to $\lambda x_0$: the possible failure of Littlewood's bound, if any, is visible within a bounded multiplicative distance of the start of an excursion. Quantitatively:

**Corollary 8.14** (Two-sided localization of large values)**.** *Suppose* $\Theta > 1/2$*, and fix* $0 < \eta < \Theta - \frac{1}{2}$ *and* $\lambda > \lambda_*$*. Among the excursions starting at* $x_0 \le X$*, at least*

$$\Big(\frac{\gamma_1}{\pi} - \frac{1}{\log\lambda} - o(1)\Big)\log X$$

*contain a point* $t_*$ *with* $|M(t_*)| \ge \frac{1}{2}t_*^{\Theta-\eta}$ *and*

$$x_0 + \tfrac{1}{2}x_0^{\Theta-\eta} \le t_* \le \lambda x_0.$$

*Proof.* Of the excursions of Corollary 8.7, those whose witness exceeds $\lambda x_0$ are $\lambda$-long, hence at most $\log X/\log\lambda + 2$ by Lemma 8.4. For the others, $M(x_0) = 0$ and unit steps give $|M(t_*)| \le t_* - x_0$, so $t_* - x_0 \ge \frac{1}{2}t_*^{\Theta-\eta} \ge \frac{1}{2}x_0^{\Theta-\eta}$. □

*Remark* 8.15 (The threshold $\lambda_*$). The function $\cos(\gamma_1\log x)$ changes sign $\gamma_1/\pi$ times per unit of $\log x$, i.e. once in every multiplicative interval of ratio $e^{\pi/\gamma_1} = \lambda_*$; the constant in (8.5) is this density. On the geometric side, a chain of $\lambda$-long intervals realizes the bound of Lemma 8.4. The method of Theorem 8.11 compares these two counts, and $\lambda_*$ is the exact point where they balance: for $\lambda \le \lambda_*$ the $\lambda$-long family can, as far as counting alone is concerned, absorb every Kaczorowski–Pintz excursion. Going below $\lambda_*$ requires information on where inside an excursion the large values occur, not only how many excursions carry them.

8.4. **The linear window and the one-step recursion.** Once the horizon is below $2x_0$, the square-range bilinear structure of Proposition 8.29 is no longer needed: on $(u, 2u]$ the Möbius function is a *linear* expression in the prefix $\mu\mathbf{1}_{[1,u]}$.

**Proposition 8.16** (Linear window)**.** *Let* $u \ge 1$*. For every integer* $u < n \le 2u$*,*

$$\mu(n) = -\sum_{\substack{d\mid n\\ d\le u}} \mu(d). \tag{8.10}$$

*Consequently, if* $M(u) = 0$ *and* $u < t \le 2u$*,*

$$M(t) = -\sum_{d\le u} \mu(d)\left(\Big\lfloor\frac{t}{d}\Big\rfloor - \Big\lfloor\frac{u}{d}\Big\rfloor\right). \tag{8.11}$$

*Proof.* Since $n \ge 2$, $\sum_{d\mid n}\mu(d) = 0$. Every proper divisor of $n$ is at most $n/2 \le u$, and every divisor $d \le u$ is proper because $n > u$. Summing (8.10) over $u < n \le t$, the number of multiples of $d$ in $(u, t]$ is $\lfloor t/d\rfloor - \lfloor u/d\rfloor$. □

In convolution form, (8.10) says that $(\mu - \mu_u) * 1 = \mu\mathbf{1}_{(u,\infty)}$ on $n \le 2u$, the order-one analogue of (8.20), which for $u < n \le u^2$ reads $\mu(n) = -(\mu_u * \mu_u * 1)(n)$; on $(u, 2u]$ both give $\mu(n)$, the bilinear form being the quadratic continuation of the linear one. Equivalently, (8.11) follows from $\sum_{d\le t}\mu(d)\lfloor t/d\rfloor = 1$, since $\lfloor t/d\rfloor = 1$ for $u < d \le t \le 2u$.

**Proposition 8.17** (One-step recursion)**.** *For real $x \geq 1$,*

$$\sum_{k \leq x} M\Big(\frac{x}{k}\Big) = 1. \tag{8.12}$$

*Hence, if $M(u) = 0$ and $t > u$,*

$$M(t) = -\sum_{2 \leq k \leq t} \Big[ M\Big(\frac{t}{k}\Big) - M\Big(\frac{u}{k}\Big) \Big], \tag{8.13}$$

*with $M(y) = 0$ for $y < 1$. Every argument on the right is at most $t/2$; for $u < t \leq \frac{3}{2}u$ all arguments are at most $\frac{3}{4}u$.*

*Proof.* $\sum_{k \leq x} M(x/k) = \sum_{k \leq x} \sum_{d \leq x/k} \mu(d) = \sum_{m \leq x} \sum_{d \mid m} \mu(d) = 1$. Subtract the identity at $u$, where $M(u) = 0$, and note $M(u/k) = 0$ for $k > u$. □

*Remark* 8.18 (What the recursion determines). Iterating (8.12) determines all of $M$ from $M(1) = 1$, so the statement that $M$ on $(u, \frac{3}{2}u]$ is determined by $M$ on $[1, \frac{3}{4}u]$ carries no information by itself; cf. Remark 9.2. The content of (8.13) is the explicit one-step form: at the start $u$ of an excursion, every value in the only window that must be controlled (Corollary 8.19) is a signed sum of increments of $M$ over the scaled intervals $(u/k, t/k]$, all lying at scales at most $\frac{3}{4}u$.

**Corollary 8.19** (RH in the linear window)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$\Big| \sum_{d \leq u} \mu(d) \Big( \Big\lfloor \frac{t}{d} \Big\rfloor - \Big\lfloor \frac{u}{d} \Big\rfloor \Big) \Big| = \Big| \sum_{2 \leq k \leq t} \Big[ M\Big(\frac{t}{k}\Big) - M\Big(\frac{u}{k}\Big) \Big] \Big| \ll_\varepsilon t^{1/2+\varepsilon}$$

*for every excursion starting at $u$ and every $u < t \leq \min\{x_1, \frac{3}{2}u\}$.*

*Proof.* Corollary 8.13(iii) with $\lambda = \frac{3}{2} > \lambda_*$, and Propositions 8.16 and 8.17. □

**Proposition 8.20** (Kernel form of the recursion)**.** *Let $M(u) = 0$ and let $t$ be an integer with $u < t < 2u$. For $1 \leq q \leq t/2$ put*

$$c_q(t,u) = \Big( \Big\lfloor \frac{t}{q} \Big\rfloor - \Big\lfloor \frac{t}{q+1} \Big\rfloor \Big) - \Big( \Big\lfloor \frac{u}{q} \Big\rfloor - \Big\lfloor \frac{u}{q+1} \Big\rfloor \Big).$$

*Then:*

(i) $M(t) = -\sum_{1 \leq q \leq t/2} c_q(t,u) M(q)$;

(ii) $c_q(t,u) = \dfrac{t-u}{q(q+1)} + \rho_q$ *with* $|\rho_q| < 2$; *for* $q > \sqrt{t}$ *one has* $c_q \in \{-1, 0, 1\}$, *and* $c_q \neq 0$ *only if* $q = \lfloor t/k \rfloor$ *or* $q = \lfloor u/k \rfloor$ *for some* $k < \sqrt{t}$;

(iii) $c_1(t,u) \geq \frac{1}{2}(t-u-1)$;

(iv) $M(t) = -(t-u)\,\Theta(\sqrt{t}) + E(t,u)$, *where* $\Theta$ *is the coefficient* (10.10) *and*

$$|E(t,u)| \leq 2 \sum_{q \leq \sqrt{t}} |M(q)| + \sum_{k < \sqrt{t}} \Big( \Big| M\Big(\frac{t}{k}\Big) \Big| + \Big| M\Big(\frac{u}{k}\Big) \Big| \Big).$$

*Proof.* (i) In (8.13) group the indices $k \geq 2$ by the value $q = \lfloor t/k \rfloor$, respectively $q = \lfloor u/k \rfloor$. The number of $k \geq 1$ with $\lfloor t/k \rfloor = q$ is $\lfloor t/q \rfloor - \lfloor t/(q+1) \rfloor$, and for $q \leq t/2$ none of them is $k = 1$. The same holds for $u$, because $k = 1$ gives $q = u > t/2$. (ii) Each bracket equals $x/(q(q+1))$ minus a difference of two fractional parts, so it differs from $x/(q(q+1))$ by less than 1. For $q > \sqrt{t} \geq \sqrt{u}$ both $t/q - t/(q+1)$ and $u/q - u/(q+1)$ are $< 1$, so each bracket is 0 or 1, and it is 1 only if $q = \lfloor x/k \rfloor$ with $k \leq x/q < \sqrt{t}$. (iii) $c_1 = (t-u) - (\lfloor t/2 \rfloor - \lfloor u/2 \rfloor)$ and $\lfloor t/2 \rfloor - \lfloor u/2 \rfloor \leq \lceil (t-u)/2 \rceil$. (iv) Insert (ii) into (i) for $q \leq \sqrt{t}$; the smooth part is $(t-u)\sum_{q \leq \sqrt{t}} M(q)/(q(q+1)) = (t-u)\Theta(\sqrt{t})$. For $q > \sqrt{t}$ use the last assertion of (ii). □

**Corollary 8.21** (Loss of the pointwise induction)**.** *Let $0 < \beta < 1$ and suppose $|M(q)| \le Cq^\beta$ for all $q \le t/2$. Then, for $u < t < 2u$ with $M(u) = 0$,*

$$|M(t) + (t-u)\Theta(\sqrt{t})| \ll_\beta C\, t^{(1+\beta)/2}, \tag{8.14}$$

*while the termwise bound of the full kernel gives only $|M(t)| \ll_\beta C\, t$. For $\beta = \frac{1}{2} + \varepsilon$ the pointwise induction therefore loses $t^{1/2-\varepsilon}$, and $t^{1/4-\varepsilon/2}$ after the smooth component $-(t-u)\Theta(\sqrt{t})$ is extracted.*

*Proof.* By Proposition 8.20(iv), $|E| \le 2C\sum_{q\le\sqrt{t}} q^\beta + 2Ct^\beta \sum_{k<\sqrt{t}} k^{-\beta} \ll_\beta Ct^{(1+\beta)/2}$. The full kernel satisfies $\sum_q |c_q| q^\beta \ge c_1 \ge \frac{1}{2}(t-u-1)$ by (iii), and $\sum_q |c_q| q^\beta \ll_\beta t$ by (ii). □

The smooth component is not small for termwise reasons: by (10.12),

$$\Theta(\sqrt{t}) = -\sum_{n>\sqrt{t}} \frac{M(n)}{n(n+1)},$$

so its size is governed by the global identity $\sum_{n\ge1} M(n)/(n(n+1)) = 0$, i.e. by the pole of $\zeta$ at $s = 1$ (Remark 10.14), and involves values of $M$ beyond $t/2$. Even under RH the tail gives only $(t-u)\Theta(\sqrt{t}) \ll t^{3/4+\varepsilon/2}$, the same order as the bound for $E(t,u)$; the RH scale for $M(t)$ is a statement about their sum.

*Remark* 8.22 (The sign cone of an excursion). If $(u, x_1]$ is a positive excursion, Proposition 8.20(i) gives the system

$$\sum_{q\le(u+r)/2} c_q(u+r,u)\, M(q) < 0 \qquad (1 \le r < \min\{x_1 - u, u\}),$$

so the vector $(M(q))_{q<u}$ lies in an intersection of open half-spaces determined by $u$ and the excursion length. This constraint is used by none of the bounds above.

**Proposition 8.23** (Termwise cone contraction is empty)**.** *Let $(u, x_1]$ be an excursion, $1 \le h < \min\{x_1 - u, u\}$, and let $w_1, \dots, w_h \ge 0$, not all zero. Put $K_q = \sum_{r\le h} w_r c_q(u+r,u)$, so that $\sum_r w_r M(u+r) = -\sum_q K_q M(q)$. If, for some $0 < \beta < 1$ and $C_\beta > 0$,*

$$\sum_q |K_q|\, q^\beta \le C_\beta \sum_{r\le h} w_r (u+r)^\beta, \tag{8.15}$$

*then*

$$\sum_{r\le h} w_r |M(u+r)| \le (1 + 2C_\beta) \sum_{r\le h} w_r (u+r)^\beta.$$

*Proof.* All $c_1(u+r,u) \ge \frac{1}{2}(r-1) \ge 0$ by Proposition 8.20(iii), so $\frac{1}{2}\sum_r w_r(r-1) \le K_1 \le C_\beta \sum_r w_r (u+r)^\beta$. Since $M(u) = 0$ and $M$ has unit steps, $|M(u+r)| \le r$, and $\sum_r w_r r \le \sum_r w_r + 2C_\beta \sum_r w_r (u+r)^\beta$. □

Thus any weighting satisfying (8.15) lives where the unit-step bound alone already keeps the weighted normalized height below $1 + 2C_\beta$; since $|M(1)| = 1$, the "descent" inequality $\sum_r w_r |M(u+r)| \le C_\beta \max_{q<u}(|M(q)|/q^\beta) \sum_r w_r (u+r)^\beta$ it would yield already holds, with $1 + 2C_\beta$ in place of $C_\beta$, for trivial reasons, and forces nothing. A contraction inequality can carry information only after the smooth part $(\sum_r w_r r)/(q(q+1))$ is removed from $K_q$, with the corresponding term $\sum_r w_r r\, \Theta(\sqrt{u+r})$ controlled separately.

8.5. **Conditional extreme-value consequence of excursion sparsity.** The counts of Section 8.1 bound the number of excursions in a family, not their heights. They acquire a second function when combined with a tail hypothesis on a normalized excursion statistic: they fix the scale of the family maximum, with no independence assumption.

Let $(\Omega, \mathcal{G}, \Pr)$ be a probability space carrying a random sequence of disjoint integer intervals $I_k = (x_{0,k}, x_{1,k}]$ with $x_{1,k} \le x_{0,k+1}$, and real random variables $Z_k$ (for instance a model of the Möbius walk with $Z_k = H_k/\sqrt{\ell_k}$). A family $\mathcal{F}$ is given by a rule selecting indices from $(x_{0,k}, x_{1,k})_k$, such as $\mathcal{F}_h$; let $k_{\mathcal{F}}(1) < k_{\mathcal{F}}(2) < \cdots$ enumerate its members, and put

$$Z_{\max,\mathcal{F}}(X) := \max\{Z_k : k \in \mathcal{F},\ x_{0,k} \le X\} \qquad (\max \emptyset = -\infty).$$

The lemmas of Section 8.1 hold for every $\omega$.

**Hypothesis 8.24** ($\mathrm{SG}_p$)**.** There are $C, c, p > 0$ such that for every $j \ge 1$ and $z \ge 0$

$$\Pr\big(k_{\mathcal{F}}(j) \text{ exists and } Z_{k_{\mathcal{F}}(j)} > z\big) \le Ce^{-cz^p}.$$

For $p = 2$ this is a uniform sub-Gaussian tail, denoted (SG).

**Theorem 8.25** (Extreme scale from a count)**.** *Assume* ($\mathrm{SG}_p$) *for $\mathcal{F}$ and suppose that $E_{\mathcal{F}}(X) \le L(X)$ for every $\omega$, with $L(X) \ge 1$ deterministic. Then*

$$\Pr\big(Z_{\max,\mathcal{F}}(X) > z\big) \le L(X)\, Ce^{-cz^p}. \tag{8.16}$$

*Consequently, for every $\delta > 0$,*

$$\Pr\Big(Z_{\max,\mathcal{F}}(X) > \Big(\frac{1+\delta}{c}\log L(X)\Big)^{1/p}\Big) \le C\, L(X)^{-\delta}, \tag{8.17}$$

*and $Z_{\max,\mathcal{F}}(X) = O_{\Pr}\big((\log L(X))^{1/p}\big)$ as $L(X) \to \infty$.*

*Proof.* The members of $\mathcal{F}$ starting at most $X$ are the first $E_{\mathcal{F}}(X) \le L(X)$ members of the enumeration, so $\{Z_{\max,\mathcal{F}}(X) > z\} \subset \bigcup_{j \le L(X)} \{k_{\mathcal{F}}(j) \text{ exists},\ Z_{k_{\mathcal{F}}(j)} > z\}$, and (8.16) is the union bound. Substituting $cz^p = (1+\delta)\log L(X)$ gives (8.17). □

**Corollary 8.26** (Triple-logarithmic scale on the square family)**.** *Assume (SG) for the square-exceptional family $\mathcal{F}_{x^2}$ of the model. Then, for $X \ge e^e$ and every $\delta > 0$,*

$$\Pr\big(Z_{\max,\square}(X) > z\big) \le 4C(\log\log X)e^{-cz^2}, \tag{8.18}$$

$$\Pr\Big(Z_{\max,\square}(X) > \sqrt{\tfrac{1+\delta}{c}\log\log\log X}\Big) \le 4C(\log\log X)^{-\delta}. \tag{8.19}$$

*Thus $Z_{\max,\square}(X) = O_{\Pr}(\sqrt{\log\log\log X})$. The same holds for the family $\mathcal{F}_{cx^2}$ of Lemma 8.2 with a constant depending on $c$.*

*Proof.* Use (8.2) in (8.16) and note that $e^{-cz^2} = (\log\log X)^{-(1+\delta)}$ at the stated $z$. □

The union bound gives convergence in probability; a Borel–Cantelli argument along a sequence adapted to $L$ gives an almost sure bound with the same constant.

**Theorem 8.27** (Almost sure extreme scale)**.** *Assume* ($\mathrm{SG}_p$) *for $\mathcal{F}$ and $E_{\mathcal{F}}(X) \le L(X)$ for every $\omega$, where $L$ is continuous, nondecreasing and unbounded. Then almost surely*

$$\limsup_{X\to\infty} \frac{Z_{\max,\mathcal{F}}(X)}{(\log L(X))^{1/p}} \le c^{-1/p}.$$

*Proof.* Fix $\delta > 0$. For $j \ge j_0$ let $X_j$ be the least $X$ with $L(X) = 2^j$. Put $z_j = \big((1+\delta)(j+1)\log 2/c\big)^{1/p}$ and $A_j = \{Z_{\max,\mathcal{F}}(X_{j+1}) > z_j\}$. By (8.16), $\Pr(A_j) \le 2^{j+1}Ce^{-(1+\delta)(j+1)\log 2} = C2^{-(j+1)\delta}$, which is summable, so almost surely $A_j$ occurs only finitely often. For $X_j \le X \le X_{j+1}$, monotonicity of $Z_{\max,\mathcal{F}}$ in $X$ gives $Z_{\max,\mathcal{F}}(X) \le z_j$ outside $A_j$, while $\log L(X) \ge j\log 2$. Hence

the lim sup is at most $\lim_j((1+\delta)(j+1)/(cj))^{1/p} = ((1+\delta)/c)^{1/p}$. Let $\delta \downarrow 0$ along a countable sequence. □

With $L_\square(X) = 2 + \log_2(\log X/\log 2)$, which is continuous and increasing, Theorem 8.27 gives under (SG)

$$\limsup_{X\to\infty} \frac{Z_{\max,\square}(X)}{\sqrt{\log\log\log X}} \le c^{-1/2} \quad \text{almost surely.}$$

Applied to the three counts available for any sequence of disjoint intervals (all excursions: at most $X$; Lemma 8.4; Lemma 8.1), Theorems 8.25 and 8.27 give three levels:

| family | members starting $\le X$ | extreme scale under (SG) |
|---|---|---|
| all excursions | $\le X$ | $\sqrt{\log X}$ |
| $x_1 > \lambda x_0$ | $O(\log X)$ | $\sqrt{\log\log X}$ |
| $x_1 > c x_0^2$ (or $x_0^A$) | $O(\log\log X)$ | $\sqrt{\log\log\log X}$ |

Each level of lacunarity removes one logarithm from the extreme scale. Under ($\mathrm{SG}_p$) the last column reads $(\log X)^{1/p}$, $(\log\log X)^{1/p}$, $(\log\log\log X)^{1/p}$.

*Remark* 8.28 (Scope). For $M$ itself the square-exceptional family is finite by Corollary 8.8, so Theorems 8.25–8.27 concern stochastic models of the Möbius walk, in which the pathwise counts of Section 8.1 persist while no analogue of the Kaczorowski–Pintz localization is available. Hypothesis (SG) is an assumption on the model; the theorems state exactly what the counts contribute to its extremes.

### 8.6. **The square range as an exact bilinear form of the prefix.**

**Proposition 8.29** (Bilinear square-range identity)**.** *Let $u \ge 1$ and $1 \le t \le u^2$. Then*

$$M(t) = 2M(\min(t,u)) - \sum_{d,j\le u} \mu(d)\mu(j) \left\lfloor \frac{t}{dj} \right\rfloor. \tag{8.20}$$

*In particular, if $M(u) = 0$ and $u < t \le u^2$,*

$$M(t) = -\mathcal{B}_u(t), \qquad \mathcal{B}_u(t) := \sum_{d,j\le u} \mu(d)\mu(j) \left\lfloor \frac{t}{dj} \right\rfloor, \qquad \mathcal{B}_u(u) = 0. \tag{8.21}$$

*Proof.* Let $\mu_u = \mu \mathbf{1}_{[1,u]}$. Each factor of $(\mu - \mu_u) * (\mu - \mu_u)$ is supported on integers $> u$, so the convolution, and its convolution with 1, vanish on $n \le u^2$. Expanding and using $\mu * 1 = \delta$ gives $\mu(n) = 2\mu_u(n) - (\mu_u * \mu_u * 1)(n)$ for $n \le u^2$; sum over $n \le t$. □

*Remark* 8.30 (Higher-order determinacy). For every $k \ge 1$ and $n \le u^k$,

$$\mu(n) = \sum_{j=1}^{k} (-1)^{j-1} \binom{k}{j} (\mu_u^{*j} * 1^{*(j-1)})(n), \tag{8.22}$$

because $(\mu - \mu_u)^{*k} * 1^{*(k-1)}$ vanishes on $n \le u^k$; expand by the binomial theorem and use $\mu * 1 = \delta$. Thus determination by the prefix is not special to $u^2$. What is special to $t \le u^2$ is the order-two form (8.21), with rank-one coefficients $\mu(d)\mu(j)$ and elementary kernel $\lfloor t/(dj) \rfloor$. This is a truncation identity of Heath-Brown/Vaughan type [4]; summed forms of this kind are used in the fast computation of $M$ [5].

**Question 8.31** (Square-excursion bridges)**.** Let $u = x_{0,\nu}$ and $x_{1,\nu} \le u^2$. Then $\mathcal{B}_u(u) = \mathcal{B}_u(x_{1,\nu}) = 0$ and $-\sigma_\nu \mathcal{B}_u(t) > 0$ for $u < t < x_{1,\nu}$. Can the rank-one coefficients $\mu(d)\mu(j)$, the kernel $\lfloor t/(dj) \rfloor$ and these bridge and sign constraints be combined to prove

$$\max_{u<t\le x_{1,\nu}} |\mathcal{B}_u(t)| \ll_\varepsilon x_{1,\nu}^{1/2+\varepsilon}$$

uniformly over square-ordinary excursions? By Corollary 8.12 this uniform bound is equivalent to RH. By Corollary 8.19 it suffices to consider $t \le \min\{x_{1,\nu}, \frac{3}{2}u\}$, where $\mathcal{B}_u(t)$ reduces to the linear form (8.11).

8.7. **Mellin mass of a complete excursion.**

**Proposition 8.32** (Excursion blocks of $1/\zeta$)**.** *For a complete excursion $I = (x_0, x_1]$ put $D_I(s) = \sum_{x_0<n\le x_1} \mu(n)n^{-s}$. Then*

$$D_I(s) = \sum_{n=x_0+1}^{x_1-1} M(n)(n^{-s} - (n+1)^{-s}). \tag{8.23}$$

*For real $\sigma > 0$, $D_I(\sigma)$ has the sign of the excursion and*

$$|D_I(\sigma)| \ge \sigma\beta_I x_1^{-\sigma-1} \ge \sigma H_I^2 x_1^{-\sigma-1}. \tag{8.24}$$

*For $\Re s > 1$,*

$$\frac{1}{\zeta(s)} = \sum_I D_I(s), \tag{8.25}$$

*zero runs contributing nothing.*

*Proof.* Summation by parts gives (8.23) because $M$ vanishes at both endpoints. All interior $M(n)$ have the sign of the excursion, and $n^{-\sigma} - (n+1)^{-\sigma} \ge \sigma x_1^{-\sigma-1}$ for $n < x_1$. The bound $\beta_I \ge H_I^2$ is (4.13). If $M$ vanishes at two consecutive integers the intervening $\mu(n)$ is zero, so every $n$ with $\mu(n) \ne 0$ lies in an excursion; sum the disjoint blocks in the half-plane of absolute convergence. □

## 9. Trial-division horizon of an excursion

**Lemma 9.1** (Square horizon)**.** *Let $u \ge 2$. Trial division by the primes $p \le u$ alone determines $\mu(n)$ for every $n \le u^2$.*

*Proof.* Remove the prime factors $p \le u$ of $n$; if one occurs twice, $\mu(n) = 0$. The cofactor $q \le u^2$ has all prime factors $> u$, so it is 1 or a single prime. □

*Remark* 9.2 (Determinacy is not information). Primes $> u$ are not logically new information, and determination by a finite prefix is not special to $u^2$: (8.22) propagates a prefix to every power range, and (8.12) determines $M$ from $M(1)$. The ranges $(u, 2u]$ and $(u, u^2]$ are special only in the concrete sense that the exact summatory formula is of order one ((8.11)) or two ((8.21)) in the prefix, and that one-stage trial division by primes $\le u$ suffices on $(u, u^2]$.

**Definition 9.3** (Smooth and rough parts)**.** For $y > 1$ write $n = s_y(n)\, r_y(n)$, where every prime factor of $s_y(n)$ is $\le y$ and every prime factor of $r_y(n)$ is $> y$. Let $P^+(n)$ be the largest prime factor of $n$, with $P^+(1) = 1$.

**Lemma 9.4.** *$\mu(n) = \mu(s_y(n))\,\mu(r_y(n))$. If $n < y^k$ then $\Omega(r_y(n)) \le k-1$, and if $n \le y^2$ then $r_y(n)$ is 1 or a prime.*

*Proof.* The two parts are coprime. If $\Omega(r) \ge 1$ then $y^{\Omega(r)} < r \le n$. □

*Remark* 9.5 (The horizon $u^2$ is not the multiplicative threshold of an excursion). For a square-ordinary excursion $\sqrt{v_\nu} \le u_\nu$. Every $n \le v_\nu$ has at most one prime factor $> \sqrt{v_\nu}$, so the threshold that governs the multiplicative structure inside $I_\nu$ is $\sqrt{v_\nu}$, below $u_\nu$. The regime $t > u_\nu^2$ of Section 10 lies beyond $v_\nu$ for a square-ordinary excursion; that section concerns $M(t)$ globally, and Section 11 gives the local version inside an excursion.

## 10. Large-prime decomposition at level $u^2$

The identities of this section are decompositions by large prime factors of Buchstab/Legendre type. The asymptotic statements use only the prime number theorem with classical error term and the Vinogradov–Korobov bound [12, 19].

**Definition 10.1** (Smooth baseline and large-prime part)**.** For $u \geq 2$ and $t \geq 1$ put

$$S_u(t) := \sum_{\substack{n \leq t \\ P^+(n) \leq u^2}} \mu(n), \qquad E_u(t) := \sum_{\substack{n \leq t \\ P^+(n) > u^2}} \mu(n), \qquad M(t) = S_u(t) + E_u(t). \tag{10.1}$$

**Proposition 10.2** (Single-prime copy)**.** *For a prime $p$ and $t < p^2$,*

$$\Delta_p(t) := \sum_{a \leq t/p} \mu(ap) = -M(t/p). \tag{10.2}$$

*Proof.* $a \leq t/p < p$, so $p \nmid a$ and $\mu(ap) = -\mu(a)$. □

**Proposition 10.3** (Sign content of one prime cone)**.** *Let $\mathcal{S}(y) = \sum_{a \leq y} \mu(a)^2$. Among the numbers $ap$ with $a \leq y < p$, the numbers of values $+1, -1, 0$ of $\mu$ are*

$$N_+ = \frac{\mathcal{S}(y) - M(y)}{2}, \qquad N_- = \frac{\mathcal{S}(y) + M(y)}{2}, \qquad N_0 = \lfloor y \rfloor - \mathcal{S}(y), \tag{10.3}$$

*with limiting proportions $3/\pi^2,\ 3/\pi^2,\ 1 - 6/\pi^2$.*

*Proof.* $\mu(ap) = -\mu(a)$, and $\#\{a \leq y : \mu(a) = \mp 1\} = (\mathcal{S}(y) \mp M(y))/2$. □

**Proposition 10.4** (First-order window)**.** *If $u^2 < t < u^4$, no $n \leq t$ has two prime factors $> u^2$ (with multiplicity), and*

$$E_u(t) = -\sum_{u^2 < p \leq t} M(t/p). \tag{10.4}$$

*Proof.* Two such factors would give $n > u^4 > t$. Hence $E_u(t) = \sum_{u^2 < p \leq t} \Delta_p(t)$, and $t < u^4 < p^2$; apply (10.2). □

**Proposition 10.5** (Prime-count form)**.** *For $u^2 < t < u^4$ and $R := t/u^2$,*

$$E_u(t) = -\sum_{a \leq R} \mu(a)\big(\pi(t/a) - \pi(u^2)\big), \qquad M(t) = S_u(t) - \sum_{a \leq R} \mu(a)\pi(t/a) + M(R)\pi(u^2). \tag{10.5}$$

*Proof.* Expand (10.4) as $-\sum_p \sum_{a \leq t/p} \mu(a)$ and interchange: for fixed $a$ the prime runs over $u^2 < p \leq t/a$, which is non-empty only if $a < R$. □

Thus the only input from primes $> u^2$ is the value of $\pi$ at the points $t/a$, $a \leq R$, weighted by $\mu(a)$ at the small scale $R < u^2$.

**Definition 10.6** (Copy multiplicities)**.** For $u^2 < t < u^4$, $R = t/u^2$ and $1 \leq y \leq \lfloor R \rfloor$ put

$$N_y(t,u) = \#\Big\{p :\ \max\big(u^2, \tfrac{t}{y+1}\big) < p \leq \tfrac{t}{y}\Big\}, \qquad \text{so that} \qquad E_u(t) = -\sum_{1 \leq y \leq \lfloor R \rfloor} N_y(t,u) M(y). \tag{10.6}$$

*Remark* 10.7 (Asymptotics of the multiplicities). For fixed $R > 1$ and $u \to \infty$,

$$N_y \sim \frac{Ru^2}{2\log u} \cdot \frac{1}{y(y+1)} \quad (1 \leq y \leq \lfloor R \rfloor - 1), \qquad N_{\lfloor R \rfloor} \sim \frac{u^2}{2\log u}\Big(\frac{R}{\lfloor R \rfloor} - 1\Big) \quad (R \notin \mathbb{Z}), \tag{10.7}$$

and $N_R = 0$ if $R \in \mathbb{Z}$. The top multiplicity is truncated because $t/(\lfloor R \rfloor + 1) < u^2$.

**Definition 10.8** (Coherent coefficient)**.**

$$\mathcal{C}(R) := \int_1^R \frac{M(w)}{w^2}\,dw = \sum_{a\le R}\mu(a)\Big(\frac{1}{a}-\frac{1}{R}\Big) = \sum_{a\le R}\frac{\mu(a)}{a} - \frac{M(R)}{R}. \tag{10.8}$$

**Proposition 10.9** (Fixed $R$)**.** *For fixed $R>1$, as $u\to\infty$,*

$$E_u(Ru^2) = -\frac{Ru^2}{2\log u}(\mathcal{C}(R)+o(1)). \tag{10.9}$$

*Proof.* In (10.5) the sum is finite; for $a<R$ the prime number theorem gives $\pi(Ru^2/a)-\pi(u^2)\sim (R/a-1)u^2/(2\log u)$, and $\sum_{a\le R}\mu(a)(R/a-1)=R\,\mathcal{C}(R)$. □

**Lemma 10.10** (The multiplicity-weighted coefficient)**.** *The coefficient*

$$\Theta(R) := \sum_{1\le y\le R}\frac{M(y)}{y(y+1)} = \sum_{n\le R}\frac{\mu(n)}{n} - \frac{M(R)}{\lfloor R\rfloor+1} \tag{10.10}$$

*satisfies*

$$\Theta(R) = \mathcal{C}(\lfloor R\rfloor+1), \qquad \mathcal{C}(R) = \Theta(R) - M(R)\Big(\frac{1}{R}-\frac{1}{\lfloor R\rfloor+1}\Big), \tag{10.11}$$

*and*

$$\mathcal{C}(R) = -\int_R^\infty \frac{M(w)}{w^2}\,dw, \qquad \Theta(R) = -\sum_{n>R}\frac{M(n)}{n(n+1)}. \tag{10.12}$$

*Proof.* $M$ is constant on $[y,y+1)$ and $\int_y^{y+1} w^{-2}dw = 1/(y(y+1))$, which gives (10.11). For $s>1$, (7.1) gives $\int_1^\infty M(w)w^{-s-1}\,dw = 1/(s\zeta(s))\to 0$ as $s\downarrow 1$. For $s\in[1,1+\delta]$ the bound $M(w)\ll we^{-c\sqrt{\log w}}$ dominates the integrand by an integrable multiple of $e^{-c\sqrt{\log w}}/w$, so dominated convergence gives $\int_1^\infty M(w)w^{-2}dw=0$. □

*Remark* 10.11 ($\mathcal{C}$ versus $\Theta$). Summing (10.7) against $M(y)$ suggests the coefficient $\Theta(R)$, but the truncated top multiplicity replaces it by $\mathcal{C}(R)$; by (10.11) the two agree exactly when $M(\lfloor R\rfloor)=0$. For $1<R<2$ one has $E_u=-(\pi(t)-\pi(u^2))\sim -(R-1)u^2/(2\log u)$, i.e. $\mathcal{C}(R)=1-1/R$, whereas $\Theta(R)=1/2$.

**Corollary 10.12** (Decay of the coefficients)**.** *By* (10.12) *and* (6.2)*,*

$$\mathcal{C}(R),\ \Theta(R) \ll \exp(-c_1(\log R)^{3/5}(\log\log R)^{-1/5}). \tag{10.13}$$

**Proposition 10.13** (Both parts are large)**.** (i) *For fixed $R>1$ with $\mathcal{C}(R)\neq 0$,*

$$E_u(Ru^2)\sim -\frac{Ru^2}{2\log u}\mathcal{C}(R), \qquad S_u(Ru^2)\sim +\frac{Ru^2}{2\log u}\mathcal{C}(R); \tag{10.14}$$

*for $1<R<2$, $\mathcal{C}(R)=1-1/R>0$.*

(ii) *There is $C>0$ such that, as $t\to\infty$, uniformly for $u^2<t<u^4$ with $R=t/u^2\ge \exp(C(\log\log t)^2)$,*

$$E_u(t) = (1+o(1))\frac{t}{(\log t)^2}, \qquad S_u(t) = -(1+o(1))\frac{t}{(\log t)^2}. \tag{10.15}$$

*In particular, in both regimes $|E_u(t)|$ and $|S_u(t)|$ exceed $t^{1/2+\varepsilon}$ for every fixed $\varepsilon<1/2$ and all large $t$, and only their sum $M$ can be compared with the scale $t^{1/2+\varepsilon}$.*

*Proof.* We use the classical form [12]: for some $c > 0$,

$$\pi(y) = \mathrm{li}(y) + O(ye^{-c\sqrt{\log y}}), \quad M(y) \ll ye^{-c\sqrt{\log y}}, \quad \widetilde{m}(y) := \sum_{a\le y} \frac{\mu(a)}{a} \ll e^{-c\sqrt{\log y}}, \tag{10.16}$$

and $\sum_{n\ge1} \mu(n)\log n/n = -1$ as a convergent series (the value at $s = 1$ of $-\frac{d}{ds}\zeta(s)^{-1}$, since $\zeta'/\zeta^2 \to -1$). The integral below converges absolutely by the bound on $\widetilde{m}$, and

$$\int_1^\infty \widetilde{m}(w)\frac{dw}{w} = \lim_{Y\to\infty}\Big(\widetilde{m}(Y)\log Y - \sum_{a\le Y}\frac{\mu(a)\log a}{a}\Big) = 1. \tag{10.17}$$

(i) Proposition 10.9, and $M(t) = o(t/\log t)$. For $1 < R < 2$, $M = 1$ on $[1, 2)$.

(ii) Since $t < u^4$, $R < \sqrt{t}$, so $\log(t/a) \ge \frac12 \log t$ for $a \le R$. By (10.5),

$$E_u(t) = -\sum_{a\le R}\mu(a)\,\mathrm{li}(t/a) - \sum_{a\le R}\mu(a)(\pi(t/a) - \mathrm{li}(t/a)) + M(R)\pi(u^2). \tag{10.18}$$

The second term is $\ll t(1 + \log R)e^{-c\sqrt{\log t/2}} = o(t/\log^2 t)$. For the third, $\frac14 \log t < \log u < \frac12 \log t$, so

$$|M(R)|\pi(u^2) \ll Re^{-c\sqrt{\log R}}\frac{u^2}{\log u} \ll e^{-c\sqrt{\log R}}\frac{t}{\log t} = o(t/\log^2 t)$$

once $c\sqrt{C} > 1$. For the first, with $F(w) = w\,\mathrm{li}(t/w)$,

$$\sum_{a\le R}\mu(a)\,\mathrm{li}(t/a) = \widetilde{m}(R)F(R) - \int_1^R \widetilde{m}(w)F'(w)\,dw, \tag{10.19}$$

$$F'(w) = \mathrm{li}(t/w) - \frac{t/w}{\log(t/w)} = \frac{t/w}{\log^2(t/w)}\Big(1 + O\Big(\frac{1}{\log t}\Big)\Big).$$

The boundary term is $\ll e^{-c\sqrt{\log R}}t/\log t = o(t/\log^2 t)$. The integral equals

$$\frac{t}{\log^2 t}\left(\int_1^R \widetilde{m}(w)g_t(w)\frac{dw}{w} + O(1/\log t)\right), \qquad g_t(w) = \frac{\log^2 t}{\log^2(t/w)} \in [1, 4].$$

Extend $g_t(w)\mathbf{1}_{\{w\le R\}}$ by zero to $[1,\infty)$. Since $R \ge \exp(C(\log\log t)^2) \to \infty$, this factor tends to 1 pointwise and is bounded by 4, while $|\widetilde{m}(w)|/w$ is integrable; dominated convergence gives $\int_1^R \widetilde{m}(w)g_t(w)\,dw/w = 1 + o(1)$, where the $o(1)$ depends only on $t$ and tends to 0 as $t \to \infty$ uniformly over all pairs $(u, R)$ in the stated range: indeed $g_t(w)\mathbf{1}_{\{w\le R\}} \to 1$ uniformly for $w$ in compact sets, since $R \ge \exp(C(\log\log t)^2)$ and $w \le W$ give $|g_t(w) - 1| \ll \log W/\log t$. Hence $E_u(t) = (1 + o(1))t/\log^2 t$, and $S_u = M - E_u$ with $M(t) = o(t/\log^2 t)$. □

*Remark* 10.14 (Why $t/\log^2 t$). $\mathcal{C}(R)$ captures only the $1/\log t$ term, whose coefficient tends to zero by Corollary 10.12. The $1/\log^2 t$ term comes from $\zeta'(s)/\zeta(s)^2 \to -1$, that is, from the pole of $\zeta$ at $s = 1$, and does not decay in $R$.

*Remark* 10.15 (First copy transitions). For $1 < R < 2$ every copy has length 1 and $E_u(t) = -(\pi(t) - \pi(u^2))$. For $2 < R < 3$ copies of length 2 appear but $M(2) = 0$. For $3 < R < 4$ copies of length 3 enter with $M(3) = -1$, of opposite net sign.

## 11. Local large-prime decomposition of an excursion

**Proposition 11.1** (Local decomposition)**.** *Let $I_\nu = (u, v]$ be a complete excursion and $y = \sqrt{v}$. For $u < t \le v$,*

$$M(t) = \sum_{\substack{u<n\le t\\ P^+(n)\le y}}\mu(n) - \sum_{a<y}\mu(a)\,\Pi_a(t), \qquad \Pi_a(t) := \#\Big\{p:\ \max\big(y, \tfrac{u}{a}\big) < p \le \tfrac{t}{a}\Big\}, \tag{11.1}$$

*and the bridge condition $M(v) = 0$ reads*

$$\sum_{\substack{u<n\le v\\ P^+(n)\le\sqrt{v}}} \mu(n) = \sum_{a<\sqrt{v}} \mu(a)\,\Pi_a(v). \tag{11.2}$$

*Proof.* $M(u) = 0$. An $n \le v = y^2$ has at most one prime factor $p > y$; then $n = ap$ with $a < v/y = y < p$, so $\mu(n) = -\mu(a)$, and $(a, p) \mapsto ap$ is a bijection onto these $n$. □

*Remark* 11.2 (Primes in short intervals). The decomposition holds for every excursion, with threshold $\sqrt{v_\nu}$. The large-prime input consists of prime counts in the intervals $(u/a, t/a]$, of length at most $\ell_\nu/a$ at height about $v_\nu/a$. Huxley proved the prime number theorem uniformly in intervals of length $h \ge x^{7/12+\varepsilon}$ [7]; the zero-density estimates of Guth–Maynard give intervals of length $x^{17/30+o(1)}$ [3]. For Möbius sums, Matomäki–Teräväinen prove $\sum_{x<n\le x+x^\theta} \mu(n) = o(x^\theta)$ for every $\theta > 0.55$ [11]. Below these ranges no asymptotic for $\Pi_a$ is available uniformly in $a$.

**Lemma 11.3** (Inverse logarithms)**.** *Let $C$ be as in Proposition 10.13. Uniformly for*

$$\exp\bigl(C(\log\log x)^2\bigr) \le z \le \sqrt{x},$$

$$\sum_{a\le z} \frac{\mu(a)}{a\log(x/a)} = -\frac{1+o(1)}{(\log x)^2}. \tag{11.3}$$

*Proof.* Partial summation gives

$$\frac{\widetilde{m}(z)}{\log(x/z)} - \int_1^z \frac{\widetilde{m}(w)}{w\log^2(x/w)}\,dw.$$

Since $z \le \sqrt{x}$, the boundary term is $O(e^{-c\sqrt{\log z}}/\log x) = o(1/\log^2 x)$ uniformly. Write $\log^{-2}(x/w) = g_x(w)\log^{-2} x$ with $1 \le g_x(w) \le 4$, extend $g_x\mathbf{1}_{\{w\le z\}}$ by zero, and apply dominated convergence with (10.17). □

**Heuristic 11.4** (Opposite drifts)**.** Let $u \ge \sqrt{v}\exp(C(\log\log v)^2)$. Near $t \in I_\nu$ the counts $\Pi_a$ grow for $a < t/\sqrt{v}$, at rate $1/(a\log(t/a))$ if primes near $t/a$ have their expected local density uniformly in $a$. Over $(t, t+h]$ the large-prime part then changes by about $-h\sum_{a<t/\sqrt{v}} \mu(a)/(a\log(t/a)) = (1+o(1))h/\log^2 t$ by Lemma 11.3. Since $M(v) - M(u) = 0$, the $\sqrt{v}$-smooth part drifts at rate about $-1/\log^2 v$, independently of $\sigma_\nu$. As

$$\frac{\ell_\nu}{\log^2 v_\nu} \gg \sqrt{\ell_\nu} \iff \ell_\nu \gg \log^4 v_\nu,$$

for $\ell_\nu \gg (\log v_\nu)^4$ both drift contributions exceed $\sqrt{\ell_\nu}$, and (11.2) is a balance between two opposite drifts: the local analogue of Proposition 10.13.

## 12. A one-frequency model for the black curve

**Heuristic 12.1** (Scale and period)**.** Motivated by (5.1), model $M(x) \approx \sqrt{x}\,F(\log x)$ with $F$ almost periodic, and retain only the lowest zero $\gamma_1$ (RH does not imply that its coefficient dominates). Within an excursion of length $\asymp x$ this gives $H_\nu \asymp \sqrt{x}$ and $\sqrt{H_\nu}\ell_\nu \asymp x^{5/4}$, so the black curve oscillates at scale

$$|\mathcal{A}(x)| \asymp x^{5/4}, \tag{12.1}$$

the scale of Corollary 6.4, with period $2\pi/\gamma_1$ in $\log x$, i.e. a factor $e^{2\pi/\gamma_1} \approx 1.5597$ in $x$. In this model consecutive zeros of $M$ are separated by the factor $e^{\pi/\gamma_1} = \lambda_*$, the threshold of Corollary 8.13(iii): a model excursion is exactly $\lambda_*$-long.

## 13. Numerical data

The following census covers $n \le N = 455\,052\,511$, with $Z(N) = 87\,385$ excursions, recording $(u_\nu, \tau_\nu, v_\nu, H_\nu, \sigma_\nu)$ and the curves $\mathcal{A}_+, \mathcal{A}_-, \mathcal{A}$:

$$\mathcal{A}_+(N) = 6.3449 \times 10^9, \quad \mathcal{A}_-(N) = -7.0877 \times 10^9, \quad \mathcal{A}(N) = -7.4279 \times 10^8, \tag{13.1}$$

$$\max_\nu \frac{H_\nu}{\sqrt{\ell_\nu}} = 2.0743, \qquad \max_\nu \frac{\beta_\nu}{\ell_\nu^{3/2}} = 1.0214. \tag{13.2}$$

Since $|M(x)| \le 0.571\sqrt{x}$ for $33 \le x \le 10^{16}$ [6], and $|M(n)| \le \sqrt{n}$ for $n < 33$ with equality only at $n = 1$, the constant $c = 1$ is admissible in Lemma 6.1, which gives $\mathcal{A}_+(N) + |\mathcal{A}_-(N)| \le \frac{4}{5}N^{5/4} = 5.317 \times 10^{10}$ against the observed $1.3433 \times 10^{10}$, a ratio of 3.96. Heuristic 7.5 with $L = \sqrt{N}$ predicts $\#\{\ell_\nu > \sqrt{N}\} \asymp N^{1/4} \approx 146$. At this range $N^{0.05} \approx 2.7$, so factors $x^\varepsilon$ are not separable from constants. Reproducing these values requires the excursion file, the area-integration convention of Proposition 3.2, and the terminal convention of Definition 2.5.

## 14. Open problems

(1) Question 8.31, in the linear form of Corollary 8.19: bound the linear window sum

$$\sum_{d \le u} \mu(d)\big(\lfloor t/d \rfloor - \lfloor u/d \rfloor\big)$$

by $t^{1/2+\varepsilon}$ on $u < t \le \min\{x_1, \frac{3}{2}u\}$, using that $u$ is a zero of $M$ and that the sum has constant sign on the window.

(2) *Descent through the one-step recursion.* Fix $0 < \beta < 1$, $C > 0$, $\theta < 1$, and put $R_\beta(x) = |M(x)|/x^\beta$. A descent lemma of the form

$$R_\beta(t) > C \implies R_\beta(q) > C \text{ for some } q \le \theta t,$$

valid for all $t$ with the *same* constant $C$, would give $R_\beta \le \max(C, \max_{x \le T} R_\beta(x))$ by iteration down to a finite range $[1, T]$, hence RH for $\beta = \frac{1}{2} + \varepsilon$; given the finite range the lemma is equivalent to this conclusion, and its interest lies in the mechanism. A version losing a factor $c < 1$ at each step, $R_\beta(q) \ge cR_\beta(t)$, loses $t^{-\log(1/c)/\log(1/\theta)}$ after iteration and does not close. Corollary 8.21 and Proposition 8.23 show that neither the pointwise recursion nor a termwise contraction on the sign cone can supply such a lemma. The remaining target is a contraction inequality for the kernel of Proposition 8.20 with its smooth part removed, restricted to the cone of Remark 8.22, together with a separate control of the smooth term through $\Theta$.

(3) Can the threshold $\lambda_* = e^{\pi/\gamma_1}$ of Corollary 8.13 be lowered? By Remark 8.15 this requires information on the position of large values inside excursions, beyond the count (8.6).

(4) Whether (6.4) is equivalent to RH.

(5) Whether the pruned local condition (7.7) holds, and whether the bridge constraints (4.5), (4.6) admit a third independent relation involving the defect $\ell_\nu/2 - H_\nu \ge 0$.

(6) Establish (SG) for $Z = H/\sqrt{\ell}$ in the simple random walk model of the Möbius walk, and compute the extreme scale of Corollary 8.26 there; the Brownian-excursion limit suggests $c = 2$.

(7) Numerically: the number of excursions with $\ell_\nu > \sqrt{N}$ against $N^{1/4}$, the maxima of $H_\nu/\sqrt{\ell_\nu}$ and $\beta_\nu/\ell_\nu^{3/2}$ over them; the split (11.1) and the $\sigma_\nu$-independence of Heuristic 11.4; the frequency of touching runs and of ballistic excursions (Heuristic 7.8); and the distribution of $x_1/x_0$, to be compared with $\lambda_*$.

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